



Overview of preventive care in adults

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INTRODUCTION

Quality health care for individuals includes two fundamental elements: appropriate treatment for current illness and appropriate preventive care to attempt to lessen future health decline. Preventive health care is an important aspect of medical practice, leading to significant improvements in overall health [1].

Several issues are involved in providing effective preventive care. Clinicians need to prioritize and decide which of the many available preventive services to recommend and which to discourage to prevent harm from inappropriate tests or interventions. Clinicians also must find a way to deliver preventive services efficiently within the context of a busy clinical practice.

This topic will discuss the approach to prevention and how to prioritize preventive services. It will also summarize primary prevention and screening recommendations primarily targeting United States adults <65 years old. The recommendations are a compilation of those from individual UpToDate topics, in which the rationale and evidence base for the recommendations are provided.

Preventive care in certain populations and disease-specific screening recommendations are discussed in more detail in specific topic reviews. As examples:

- Pregnant persons. (See "[Prenatal care: Initial assessment](#)" and "[Prenatal care: Second and third trimesters](#)".)
- Gay men. (See "[Primary care of gay men and men who have sex with men](#)".)

- Patients with human immunodeficiency virus (HIV). (See ["Primary care of adults with HIV"](#).)
- Cancer survivors. (See ["Overview of cancer survivorship care for primary care and oncology providers"](#).)
- Patients with diabetes mellitus. (See ["Overview of general medical care in nonpregnant adults with diabetes mellitus"](#).)
- Older adults. (See ["Geriatric health maintenance"](#) and ["Evidence-based approach to prevention"](#).)
- Transgender men and women. (See ["Transgender women: Evaluation and management"](#) and ["Transgender men: Evaluation and management"](#) and ["Primary care of transgender individuals"](#).)

Basic epidemiologic principles of screening in adults are presented elsewhere. (See ["Evidence-based approach to prevention"](#).)

PRIORITIZING PREVENTIVE SERVICES

Preventive services should focus on priority health problems and effective interventions. Rather than performing a standardized comprehensive examination ("full physical") on all patients, clinicians should individualize screening and prevention interventions to maximize value, which encompasses the tradeoffs between benefits, harms, and costs [2,3].

We recommend referring to the United States Preventive Services Task Force (USPSTF) [website](#) for the most up-to-date screening [recommendations](#) based on systematic reviews of available evidence. We provide a review of screening, prevention, and counseling recommendations for older adults elsewhere. (See ["Geriatric health maintenance"](#).)

Factors to consider — Priority services may differ from patient to patient depending on age, sex, and other risk factors. The effects of recommended preventive interventions on longevity or quality of life vary widely, and the potential beneficial effect of any one intervention on an individual patient varies with their lifestyle, medical and family history, and other risk factors [4]. Available tools to help clinicians prioritize preventive services include:

- The Partnership for Prevention, which ranks preventive services based on the clinically preventable burden of disease and the cost-effectiveness of interventions [5]. The list includes [aspirin](#) chemoprophylaxis to reduce cardiovascular events, childhood immunizations, and smoking cessation advice.
- The USPSTF provides a convenient [tool](#) to identify recommended patient-specific preventive services based on a patient's sex, age, pregnancy status, and tobacco use and

whether they are sexually active.

As with all clinical decisions, patient preferences are also of prime importance [6]. Risks and benefits should be communicated to patients in a way that they can understand, which can be challenging because many patients have a limited understanding of probabilities. A 2014 systematic review found that patients are better able to understand absolute than relative risks and that visual aids can be helpful [7]. Studies also suggest that patients tend to overestimate the benefits and underestimate the harms of interventions (including screening tests) [8].

Further, scheduled preventive interventions (eg, cancer screenings) may need to be deferred due to intercurrent individual illness or a larger public health crisis (eg, epidemic or pandemic infectious disease) [9]. In such circumstances, patients should be reassured that preventive interventions will be resumed when appropriate. Cancer screening during the coronavirus pandemic is discussed elsewhere. (See "[COVID-19: Considerations in patients with cancer](#)", [section on 'Impact of the pandemic on cancer screening'](#).)

Overview of USPSTF recommendations — The United States Preventive Services Task Force (USPSTF) publishes evidence-based recommendations for multiple clinical preventive services based on systematic reviews, each of which is graded based on a set of definitions [10,11].

- **USPSTF A and B level recommendations** – The USPSTF recommends certain screening and counseling interventions either with high certainty that the net benefit is substantial ("grade A"), with high certainty that the net benefit is moderate, or with moderate certainty that the net benefit is moderate to substantial ("grade B"). These grade A and B level recommendations should be offered to patients.
- **USPSTF C level recommendations** – Preventive services for which there is at least moderate certainty that the net benefit is small are rated as "grade C." The USPSTF recommends selectively offering or providing these services to individual patients based on professional judgment, patient preferences, and individual circumstances.
- **USPSTF D level recommendations** – Preventive services for which there is moderate or high certainty that the service has **no** net benefit or that the harms outweigh the benefits are rated as "grade D." The USPSTF recommends against these services.
- **USPSTF level I recommendations** – Preventive services for which the balance of benefits and harms cannot be determined, due to lack of sufficient evidence or poor-quality or conflicting evidence, are rated as "grade I." If the service is offered, patients should understand the uncertainty about the balance of benefits and harms.

Harms of screening — There are harms associated with screening, including anxiety produced by false-positive screening tests, harms associated with diagnostic testing after a

positive screening test, overdiagnosis of conditions that may be treated but would never have become clinically apparent, and cost [12-14]. Comorbid conditions may decrease the benefits and increase the harms associated with screening [15]. The benefits of screening may decrease as patients get older, particularly for cancer screening [16]. Cancer screening in older adults is discussed in detail elsewhere. (See "[Geriatric health maintenance](#)", [section on 'Cancer screening'](#).)

Clinicians should make sure that patients understand the potential harms prior to screening and consider comorbidities when discussing screening. Patients may weigh the potential benefits and harms of screening and, in some cases, decide against screening. This is a reasonable decision that should be accepted by clinicians. The optimal screening rate is less than 100 percent when patients are adequately counseled regarding the potential benefits and harms of screening.

Screening of patients who are unlikely to benefit from screening is common. As an example, in an observational study including 27,000 patients ≥ 65 years, 31 to 55 percent of those with ≥ 75 percent risk of nine-year mortality had received prostate, breast, cervical, or colorectal cancer screening within the previous two to five years [17].

HOW TO EFFICIENTLY DELIVER PREVENTION

Periodic "check-up" — There are no strict guidelines for the optimal frequency of periodic visits, and there is no evidence on which to base optimal frequency recommendations [18]. Annual examinations are not indicated for most younger patients, although people with chronic health issues (such as diabetes) warrant regular visits, with or without a periodic health maintenance visit, independent of age [19]. In the absence of such indications, we suggest periodic health maintenance visits every three years for adult patients ≤ 49 years without chronic conditions and annually for adults ≥ 50 years. For people who have no chronic conditions and rarely see a clinician, a periodic health evaluation visit may be the only opportunity to discuss preventive care.

Preventive counseling and services are important components of the periodic health check-up [20-22]. Preventive services are more likely to be discussed when patients have an established relationship with an identified clinician. Studies have shown that patients who were connected to a specific clinician have higher rates of preventive services received [22-25]. Patients with more visits to their primary care providers were more likely to undergo screening for colorectal and breast cancer, have lower incidence and mortality of colorectal and breast cancer, and have lower overall mortality. Periodic visits may have other benefits, such as developing the patient-clinician relationship, relieving patient "worry," and educating patients about how to access the health care system [21,26].

In a 2014 meta-analysis including six randomized trials, general practice-based health checks did not decrease mortality but improved surrogate outcomes (blood pressure control, decreases in total cholesterol and body mass index [BMI]), especially in high-risk patients [27]. In another meta-analysis including 14 randomized trials and over 180,000 participants, there was no beneficial effect of general health checks on total, cardiovascular, or cancer mortality despite increased detection of hypertension and hypercholesterolemia as well as increased use of antihypertensives [28]. However, some methodologic problems of the included trials (including the age of some trials, which predated the routine use of statins) may limit the interpretation of the results.

Discussing stopping or reducing screening — Changes to recommendations that result in a decrease in the intensity of screening or discontinuing screening for those patients with reduced likelihood of benefit may be problematic for patients. In particular, patients may be confused by these changes, especially if they view screening as something they can do to protect themselves from a feared condition [3]. How best to explain the rationale for less intensive or discontinued screening can be challenging. Although we know of no easy formula for conducting these discussions, some general suggestions may be useful:

- The idea that screening programs can result in harm is often new to patients. Patients should be educated that the primary issue underlying any screening recommendation is the balance between the likelihood of benefit and the risk of harm. A discussion of hypothetical examples that illustrate how screening may lead to unnecessary, difficult, and possibly risky diagnostic procedures (such as lung biopsy) or treatment (such as chemotherapy for cancer that would not have become clinically apparent over the patient's lifetime) may be helpful.
- One may acknowledge the patient's interest in preserving health and longevity and refocus the patient away from more intensive or longer screening toward other prevention activities that bring a higher likelihood of benefit.
- For older patients with limited life expectancy, it may also be useful to start the discussion of stopping screening some years before actually stopping. Older patients, advised over years that screening is an important part of medical care, may view stopping screening as an indication that they are no longer "worth" the expense or effort [29]. Again, hypothetical stories may be used to illustrate the reality that the probability of benefit from screening decreases and the probability of harm from overdiagnosis increases in older individuals. In addition, harm occurs soon after screening, while any benefit occurs only years after screening. For example, for colorectal cancer and breast cancer screening, a patient must live 10 years to have a 1 in 1000 probability of having their life extended and 16 years to have a 2 in 1000 probability [30]. If a patient is overdiagnosed, however, they receive unnecessary treatment (often including surgery) soon after diagnosis.

Supportive office systems — Creative approaches using office systems can assist in delivering needed preventive care [31]. As an example, practices have successfully used databases of eligible patients (eg, for flu shots, mammograms, diabetic eye examinations, etc) and automatic reminders to outreach to patients and assist in delivering routine preventive care [32]. In one study of interventions to promote screening for colorectal cancer, mailings to patients were somewhat more effective than electronic reminders to clinicians [33]. Finally, using decision aids and/or training medical assistants to discuss important preventive services before a visit with a clinician can be more efficient and effective for helping patients make informed decisions [34].

CARDIOVASCULAR DISEASE PREVENTION

Cardiovascular risk assessment — Patients aged ≥ 20 years should undergo cardiovascular risk assessment every three to five years.

Cardiovascular disease (CVD) risk can be estimated using one of several widely available tools, each with advantages and disadvantages. While all of the risk assessment tools have advantages and disadvantages, no single tool will be appropriate for all patients. We encourage clinicians to become familiar with and use a CVD risk calculator that has been locally endorsed and that has been validated for their locale and for patient-specific race and ethnic groups. (See ["Atherosclerotic cardiovascular disease risk assessment for primary prevention in adults"](#) and ["Cardiovascular disease risk assessment for primary prevention: Risk calculators"](#).)

The major risk factors for CVD that are modifiable include (see ["Overview of primary prevention of cardiovascular disease in adults"](#)):

- Diet. (See ["Healthy diet in adults"](#).)
- Smoking. (See ["Cardiovascular risk of smoking and benefits of smoking cessation"](#).)
- Hypertension. (See ["Cardiovascular risks of hypertension"](#).)
- Dyslipidemia. (See ["Low-density lipoprotein cholesterol-lowering therapy in the primary prevention of atherosclerotic cardiovascular disease"](#).)
- Obesity. (See ["Overweight and obesity in adults: Health consequences"](#), section on 'Cardiovascular'.)
- Physical activity. (See ["The benefits and risks of aerobic exercise"](#), section on 'Cardiovascular disease'.)

- Diabetes mellitus. (See ["Prevalence of and risk factors for coronary heart disease in patients with diabetes mellitus"](#), section on 'Prevalence and extent of increased risk'.)

Screening for coronary artery disease in patients with risk factors is discussed in detail elsewhere. (See ["Screening for coronary heart disease"](#).)

Hypertension — Hypertension screening is recommended for adults ≥ 18 years of age [35,36]. The optimal interval for screening for hypertension is not known. Most patients have their blood pressure checked at each primary care visit.

The 2021 United States Preventive Services Task Force (USPSTF) guidelines recommend hypertension screening every year for adults ≥ 40 years and for those who are at high risk for high blood pressure (eg, patients with mean daytime blood pressure 130 to 139/80 to 89 mmHg, those with overweight or obesity, and African American individuals) [37,38]. Adults aged 18 to 39 years without elevated blood pressure (ie, $< 130/80$ mmHg) and without CVD risk factors should be rescreened every three to five years. The USPSTF recommends obtaining out-of-office blood pressure measurements to confirm a diagnosis of hypertension before starting treatment. (See ["Hypertension in adults: Blood pressure measurement and diagnosis"](#), section on 'Screening for hypertension'.)

Recommendations for target blood pressure and treatment differ based on several factors, including patient age, comorbidities, and method of blood pressure measurement ([table 1](#)). (See ["Hypertension in adults: Blood pressure measurement and diagnosis"](#) and ["Hypertension in adults: Goal blood pressure"](#).)

Hyperlipidemia

- We suggest that patients aged 17 to 21 years undergo one-time screening for hyperlipidemia with a nonfasting non-high-density lipoprotein (non-HDL) cholesterol; non-HDL cholesterol is the difference between total cholesterol and HDL cholesterol [39]. (See ["Dyslipidemia in children and adolescents: Definition, screening, and diagnosis"](#).)
- For patients with a normal screen before age 21 who are also at high risk, we suggest screening for lipid abnormalities starting at age 25 for men and 35 for women. We consider patients at high risk if they have more than one risk factor (eg, diabetes, hypertension, smoking, family history) or a single risk factor that is severe (eg, several siblings with coronary artery disease in their 40s, a very heavy smoker).

Definition of hypertension based on blood pressure measurement strategy					
SBP/DBP	Clinic	SMBP	Daytime ABPM	Nighttime ABPM	24-hour ABPM
ACC/AHA guidelines 2017/18	$\geq 130/80$	$\geq 130/80$	$\geq 130/80$	$\geq 135/85$	$\geq 125/75$
ESC guidelines 2018/19	$\geq 140/90$	$\geq 135/85$	$\geq 135/85$	$\geq 120/70$	$\geq 130/80$

Definition of hypertension based on blood pressure measurement strategy

Table 1 - larger image below

- For patients with a normal screen before age 21 who are not at high risk, we suggest screening for lipid abnormalities starting at age 35 for men and 45 for women.

In patients whose lipid measurements place them well below the threshold for treatment (including lifestyle modification or lifestyle modification plus pharmacologic therapy), we suggest repeating measurements every five years. In patients who have had multiple lipid screenings with acceptable measurements, we suggest stopping screening at age 65. Ongoing assessment of cardiovascular risk can be based on other risk factors and earlier lipid levels. (See ["Screening for lipid disorders in adults"](#).)

When a decision is made to screen lipids to assess cardiovascular risk, we suggest measuring the total cholesterol and HDL cholesterol rather than a complete lipid profile or other lipid marker or fractions. Testing for total cholesterol and HDL cholesterol does not require a fasting test. (See ["Screening for lipid disorders in adults"](#), section on 'Choice of tests'.)

Obesity — We suggest screening all patients for obesity with a body mass index (BMI) and counseling all patients to maintain a healthy body weight ([calculator 1](#)). Selection of patients for treatment should be based on BMI and risk factor assessment. (See ["Obesity in adults: Prevalence, screening, and evaluation"](#) and ["Obesity in adults: Overview of management"](#).)

Physical activity — It has been estimated that up to one-third of CVD deaths may be preventable by a healthy diet and physical activity [40]. We suggest counseling patients about exercise to promote the maintenance of a healthy weight and physical activity for all patients. (See ["The benefits and risks of aerobic exercise"](#) and ["Exercise and fitness in the prevention of atherosclerotic cardiovascular disease"](#).)

Diabetes mellitus — For adults ages 35 to 70 years who are at increased risk for diabetes and/or have cardiovascular risk factors ([table 2](#)), we suggest screening for type 2 diabetes as part of cardiovascular risk assessment [41]. (See ["Screening for type 2 diabetes mellitus and prediabetes"](#).)

We use glycated hemoglobin (A1C) for diabetes screening because it correlates best with rates of future diabetes and diabetes complications. Alternative strategies include using fasting plasma glucose with or without simultaneous A1C testing. (See ["Screening for type 2 diabetes mellitus and prediabetes"](#), section on 'Screening tests'.)

Aspirin for primary prevention — Decisions regarding [aspirin](#) for primary prevention of CVD should be made on an individual basis. Factors to be considered are discussed in detail elsewhere. (See ["Aspirin in the primary prevention of cardiovascular disease and cancer"](#).)

Routine electrocardiogram — A routine electrocardiogram, even for a "baseline," is not indicated in asymptomatic adults at low risk for CVD. (See ["Screening for coronary heart disease"](#), section on 'Resting ECG and ambulatory ECG monitoring'.)

Nontraditional measures — Coronary artery calcium testing may be useful as a complement to CVD risk assessment in select patients (see ["Coronary artery calcium \(CAC\) scoring: Overview and clinical utilization"](#)). We do not recommend screening low-risk patients with other nontraditional measures (eg, C-reactive protein, carotid artery intima-media thickness, homocysteine, or lipoprotein[a]).

However, as discussed separately, these measures may be warranted for patients without known CVD who are at intermediate risk for CVD and for whom a more definite estimate of CVD risk would change management:

- C-reactive protein. (See ["C-reactive protein in cardiovascular disease"](#).)
 - Carotid artery intima-media thickness. (See ["Carotid intima-media thickness"](#).)
 - Homocysteine. (See ["Overview of homocysteine"](#).)
 - Lipoprotein(a). (See ["Lipoprotein\(a\)"](#).)
-

CANCER PREVENTION

General cancer prevention counseling — A number of measures can be taken to prevent cancer, including:

- Avoidance of tobacco (see ["Tobacco use"](#) below)
- Being physically active
- Maintaining a healthy weight
- Eating a diet rich in fruits, vegetables, and whole grains and low in saturated/trans fat
- Limiting alcohol consumption
- Protecting against sexually transmitted infections (including receiving human papillomavirus [HPV] vaccination)
- Avoiding excess sun exposure

General measures for cancer prevention are discussed in more detail separately. (See ["Overview of cancer prevention"](#).)

Tobacco use — Smoking tobacco products (primarily cigarettes) is the most important risk factor for the development of lung cancer and contributes to a large burden of disease, including other malignancies, cardiovascular disease (CVD), and pulmonary disease. (See ["Cigarette smoking and other possible risk factors for lung cancer"](#) and ["Cardiovascular risk of smoking and benefits of smoking cessation"](#) and ["Overview of cancer prevention"](#), section on ["Tobacco use"](#).)

We recommend that clinicians ask all patients whether they use tobacco and in what form. Smoking status should be documented; for all patients who use tobacco, clinicians should offer the advice to quit; assess the patient's interest in quitting; and, for those interested in quitting, offer the necessary tools (including behavioral intervention **and** pharmacotherapy), techniques, and follow-up [42]. (See ["Overview of smoking cessation management in adults"](#).)

CANCER SCREENING

Breast cancer — Major risk factors for breast cancer in women are age, genetic predisposition, and estrogen exposure ([table 3](#)). Evaluation for hereditary breast and ovarian cancer syndromes should be considered in all individuals with a personal or family history that is concerning. (See ["Factors that modify breast cancer risk in females"](#) and ["Screening for breast cancer: Strategies and recommendations"](#), section on 'Role of risk prediction models'.)

- Family history – All individuals at risk for hereditary breast and ovarian cancer syndromes should be identified by taking a thorough personal and family history, focusing on both biologic sides of the family ([table 4](#)). Patients with concerning family history should be referred to a genetic counselor for a formal genetic assessment. Recommendations for screening in patients with a hereditary breast and ovarian cancer syndrome are discussed in detail elsewhere. (See ["Overview of hereditary breast, ovarian, pancreatic and prostatic cancer syndromes"](#) and ["Cancer risks and management in BRCA1/2 carriers"](#).)
- Screening for average-risk individuals – Clinicians should address breast cancer screening starting at age 40, noting that guidelines differ on recommendations for patients under 50 ([table 5](#)). (See ["Screening for breast cancer: Strategies and recommendations"](#), section on 'Age 40 to 49 years'.)

The decision to perform screening mammography should be determined by individual patient risk and values through shared decision-making. It should be emphasized that the relative benefits and harms of screening change as an individual gets older. The absolute benefits, in terms of numbers of lives saved, are lower for younger compared with older people because of both decreased incidence of breast cancer and sensitivity of mammography in younger patients. (See ["Screening for breast cancer: Strategies and](#)

Risk and protective factors for developing breast cancer			
Risk factors	Risk group		Relative risk
	Low-risk	High-risk	
Genetic BRCA1/BRCA2 genes	Negative	Positive	2.0 to 7.0
Mother or sister with breast cancer	No	Yes	2.5
Age	35 to 39	70 to 74	18.0
Age at menarche	<12	>12	1.5
Age at first birth	<20	>30	1.5 to 3.0
Age at menopause	<45	>55	2.0
Use of contraceptive pills	Never	Ever/current use	1.07 to 1.2
Hormone replacement therapy (estrogen + progestin)	Never	Current	1.2
Alcohol	None	2 to 5 drinks/week	1.4
Breast density on mammography (percentile)	0	>75	1.8 to 6.0
Bone density	Lowest quartile	Highest quartile	2.7 to 3.5
History of a benign breast biopsy	No	Yes	1.7
History of atypical hyperplasia on biopsy	No	Yes	3.7
Protective factors			
Breast feeding (months)	<10	0	0.75
Parity	<5	0	0.75
Recreational exercise	Yes	No	0.70
Paracetamol/body mass index (BMI)	<20.9	>30.7	0.60
Gynaecological before age 25 years	Yes	No	0.5
Aspirin	at least once for 24 months	Never	0.70

Table 3 - larger image below

[recommendations"](#), section on 'Shared medical decision-making' and ["Screening for breast cancer: Strategies and recommendations"](#), section on 'Benefits and harms of screening'.)

All patients should receive counseling about the benefits and harms of screening and understand that mammography is associated with a substantial risk for false-positive findings and overdiagnosis. Given declining mortality rates from breast cancer after advances in treatment protocols, some people may choose to forego screening.

For those who choose to undergo screening, the ideal interval for screening mammography is not known. We suggest screening every one to two years, depending on patient preference. (See ["Screening for breast cancer: Strategies and recommendations"](#), section on 'Average risk: Screening'.)

- **Age to stop screening** – We suggest that breast cancer screening with mammography be continued as long as an individual has a life expectancy of at least 10 years.

We suggest that those being screened for breast cancer not undergo clinical breast examination.

We suggest that breast self-examination not be performed, except by those who express a desire to do so and who have received careful instruction to differentiate normal tissue from suspicious lumps. (See ["Screening for breast cancer: Evidence for effectiveness and harms"](#), section on 'Breast self-examination'.)

Cervical cancer — Cervical cancer screening is recommended for all individuals with a cervix, as discussed separately. Individuals who have had a total hysterectomy (ie, hysterectomy with removal of the cervix) for reasons other than cervical cancer or high-grade cervical cancer precursors do **not** need to be screened. (See ["Cervical cancer screening: Benefits, harms, screening methods, and patient risk groups"](#) and ["Cervical cancer screening in resource-abundant settings: How to screen average-risk patients"](#).)

Screening refers to testing in patients who have had all prior normal results. Patients who have had prior abnormal results require **surveillance**, and the recommended options and frequency differ based on the patient's previous results and underlying risk for developing cervical cancer. (See ["Cervical cancer screening in resource-abundant settings: How to screen average-risk patients"](#), section on 'Follow-up of abnormal results'.)

- **Persons at high risk** – Persons at high risk of cervical cancer include those who are immunocompromised, have HIV infection, or have experienced in utero exposure to diethylstilbestrol. Screening recommendations for these individuals are reviewed elsewhere. (See ["Screening for cervical cancer in patients with HIV infection and other immunocompromised states"](#).)
- **Persons at average risk**

- **Age to initiate screening** – For average-risk people, most screening guidelines in resource-abundant settings recommend initiating cervical cancer screening between ages 21 and 25 years, but the specific age to start screening varies. (See "[Cervical cancer screening in resource-abundant settings: How to screen average-risk patients](#)", section on 'Age to initiate screening'.)
- **Age to discontinue screening** – The age at which to discontinue screening varies based on the patient's prior cervical cancer screening results, life expectancy, and individual preferences. Guidelines in the United States recommend stopping screening after age 65 years in patients at average risk who had previously had adequate cervical cancer screening with normal results, as defined separately. (See "[Cervical cancer screening in resource-abundant settings: How to screen average-risk patients](#)", section on 'Discontinuing screening'.)
- **Type and frequency of screening** – Test options for cervical cancer screening include primary human papillomavirus (HPV) testing by the clinician or the patient (ie, "self-testing"), cervical cytology, and cotesting with cytology and HPV. The frequency of screening depends on the specific test. The selection among different testing strategies and evidence regarding their efficacy are discussed separately. (See "[Cervical cancer screening in resource-abundant settings: How to screen average-risk patients](#)", section on 'Testing method and frequency' and "[Cervical cancer screening: Benefits, harms, screening methods, and patient risk groups](#)", section on 'Screening methods'.)

Ovarian cancer — A family history is essential to identifying patients at high risk for ovarian cancer who might benefit from further evaluation. This is described in detail separately. (See "[Screening for ovarian cancer](#)", section on 'Family history'.)

Screening for ovarian cancer is not recommended for individuals at average risk. (See "[Screening for ovarian cancer](#)", section on 'Average-risk patients'.)

Colorectal cancer — The age of initiation and frequency of colorectal cancer screening vary with the risk:

- **No risk factors** – We recommend that average-risk patients aged 45 and older be screened for colorectal cancer. We suggest that screening be continued until the life expectancy for an individual patient is estimated as less than 10 years. For most patients, it is reasonable to stop screening at age 75 years or 85 years at the latest. One-time screening with colonoscopy (to age 83) or sigmoidoscopy (to age 84) is advised for adults who have never been screened for colorectal cancer. (See "[Screening for colorectal cancer: Strategies in patients at average risk](#)", section on 'Choosing a screening test' and "[Screening for colorectal cancer: Strategies in patients at average risk](#)".)

The [recommended interval](#) varies depending on the screening strategy. Patients should be advised of the risks and benefits of different screening options. The decision about which option to select should be made between the patient and clinician, weighing factors of effectiveness, safety, cost, and availability of the screening tests. (See "[Screening for colorectal cancer: Strategies in patients at average risk](#)".)

- Family history of colorectal cancer. (See "[Screening for colorectal cancer in patients with a family history of colorectal cancer or advanced polyp](#)".)
- Familial adenomatous polyposis. (See "[Familial adenomatous polyposis: Screening and management of patients and families](#)".)
- Lynch syndrome. (See "[Lynch syndrome \(hereditary nonpolyposis colorectal cancer\): Cancer screening and management](#)".)
- Peutz-Jeghers syndrome. (See "[Juvenile polyposis syndrome](#)".)
- Inflammatory bowel disease. (See "[Surveillance and management of dysplasia in patients with inflammatory bowel disease](#)".)

Lung cancer — For patients at increased risk for lung cancer, we suggest annual screening with low-dose helical computed tomography (CT). Further discussion on the appropriateness of lung cancer screening, including information on which patients are at increased risk, can be found elsewhere. (See "[Screening for lung cancer](#)", section on 'Synthesizing the available evidence' and "[Screening for lung cancer](#)", section on 'Our approach to counseling for screening'.)

Prostate cancer — Individual patient preferences for specific health outcomes are deciding factors in determining whether to screen for prostate cancer. Patients who are potential candidates for screening should be engaged in discussions or decision-making processes that inform them and evoke these preferences. (See "[Screening for prostate cancer](#)", section on 'Approach to screening'.)

Health care providers should periodically discuss prostate cancer screening with patients who are expected to live at least 10 years and are old enough to be at significant risk for prostate cancer. We suggest that discussions begin at age 50 in average-risk individuals. We suggest discussions begin at age 40 to 45 in those at high risk for prostate cancer (including African American patients; those with a family history of prostate cancer, particularly in relatives younger than age 65; and individuals who are known or likely to have the *BRCA1* or *BRCA2* mutations).

Melanoma

- We do not routinely screen patients at average risk for melanoma; however, we recommend that clinicians remain vigilant for any suspicious lesions identified in the course of a routine or sick visit and make appropriate referrals for further evaluation of all such lesions.

We suggest that individuals at high risk of melanoma have a regular full-body skin examination by a clinician with skin expertise. The optimal frequency of examination is unknown (see ["Screening for melanoma in adults and adolescents", section on 'High-risk patients'](#)). The United States Preventive Services Task Force (USPSTF) has found insufficient evidence for or against skin cancer screening [43]. Our rationale for screening high-risk patients is presented separately. (See ["Screening for melanoma in adults and adolescents", section on 'Rationale for screening'](#).)

- Surveillance of patients in melanoma-prone kindreds is described separately. (See ["Inherited susceptibility to melanoma", section on 'Surveillance in melanoma-prone kindreds'](#).)

Multicancer detection tests — Multicancer detection (MCD) tests have the goal of detecting cancer at early stages [44]. MCD tests analyze blood samples for molecular signals (eg, deoxyribonucleic acid [DNA], proteins) that are produced by cancer cells. Although they are intended to serve as adjuncts to standard cancer screening tests, MCDs can result in substantial downstream testing, which can be costly. A positive test result requires additional imaging and diagnostic work-up, particularly since MCD tests often do not specify the type of cancer or the organ(s) that are involved.

Although some MCD tests are commercially available and may detect cancer signals across a wide diversity of malignancies [45], their clinical utility for cancer screening remains uncertain [46]. Results from studies are generally not reflective of test performance in larger populations who typically undergo screening (ie, asymptomatic individuals in primary care). Most importantly, no studies have shown that MCD testing improves clinical outcomes, such as cancer-related deaths, particularly in comparison with existing screening methodologies [47,48]. Representative studies include:

- A prospective cohort study (PATHFINDER) performed MCD testing in a convenience sample of 6621 asymptomatic adults aged 50 years or older, 31 percent of whom had a known cancer predisposition or prior cancer (a high-risk population). A cancer signal was detected in 1.4 percent of participants, of whom 38 percent were diagnosed with cancer (true positives) and 62 percent were not (false positives) [49]. Of the true positives, 17 percent had recurrence of previous disease. The median time to diagnostic resolution was 57 days in true positives and longer (162 days) in false positives. Most participants had further work-up with laboratory tests and imaging, and procedures were done in 30 percent of participants with false-positive results and 82 percent of those with true-positive results. In this high-risk study population, the overall sensitivity of MCD testing was only moderate (29

percent), suggesting that MCD testing would need to occur in conjunction with standard cancer screening methods to be effective [48].

- A multicenter cohort study of 6238 participants in England and Wales evaluated the utility of an MCD test in adult patients who were referred for urgent investigation of nonspecific symptoms that were possibly caused by lung, gastrointestinal, or gynecologic cancers [50]. This study population was therefore not typical of an asymptomatic screening population. In this cohort of symptomatic patients, the test was highly specific (98 percent) but only moderately sensitive (66 percent). Test sensitivity was lower for early-stage cancers (24 percent for stage I cancers), and only 43 percent of all cancers diagnosed were stages I to II.

Given the limitations of MCD tests, no national organization has recommended their use for cancer screening.

OTHER SCREENING

- **Iron deficiency** – We do not routinely screen every adult for iron deficiency, but we do screen those at higher risk (eg, premenopausal women, particularly those with prior pregnancies or heavy menstrual periods, as well as individuals with conditions that might cause blood loss or iron malabsorption). (See "[Determining the cause of iron deficiency in adolescents and adults](#)", section on 'Causes (organized by mechanism)').)

The frequency of screening is also individualized; annual screening may be reasonable for those at the highest risk, such as a menstruating individual with heavy periods. Screening for iron deficiency is discussed in detail elsewhere. (See "[Diagnosis of iron deficiency and iron deficiency anemia in adults](#)", section on 'Screening (asymptomatic individuals)').)

Screening for iron deficiency in pregnant individuals is discussed elsewhere. (See "[Anemia in pregnancy](#)", section on 'Screening during pregnancy').)

- **Hypothyroidism** – Although subclinical hypothyroidism is common in the adult population [51], we do not routinely screen most asymptomatic nonpregnant adults, as there is no evidence that early detection and treatment with T4 improves clinically important outcomes in such individuals. However, we do screen individuals who are at increased risk for hypothyroidism, including those with goiter, history of autoimmune disease, previous radioactive iodine therapy and/or head and neck irradiation, family history of thyroid disease, and use of medications that may impair thyroid function. This is discussed in detail elsewhere. (See "[Diagnosis of and screening for hypothyroidism in nonpregnant adults](#)", section on 'Screening in the absence of signs and/or symptoms of hypothyroidism').)

Recommendations for screening pregnant individuals for hypothyroidism are discussed separately. (See ["Hypothyroidism during pregnancy: Clinical manifestations, diagnosis, and treatment"](#), section on 'Screening'.)

- **Vitamin D deficiency** – There are few data regarding screening for vitamin D deficiency in adults, but most experts agree that it is not necessary to perform broad-based screening of serum 25(OH)D levels in the general population. We do, however, screen those at high risk of vitamin D deficiency (eg, limited or no sunlight exposure; individuals with obesity, osteoporosis, or malabsorption). Screening for vitamin D deficiency and vitamin D replacement therapy in older adults are reviewed in detail elsewhere. (See ["Vitamin D deficiency in adults: Definition, clinical manifestations, and treatment"](#), section on 'Candidates for 25(OH)D measurements' and ["Geriatric health maintenance"](#), section on 'Vitamin D'.)
- **Glaucoma** – Although there is limited evidence to support screening, in populations with sufficient resources, we suggest that individuals over age 40 undergo periodic comprehensive eye evaluations by an eye specialist to evaluate for glaucoma. Earlier evaluation may be appropriate for individuals with significant risk factors. This is discussed in detail elsewhere. (See ["Open-angle glaucoma: Epidemiology, clinical presentation, and diagnosis"](#), section on 'Screening'.)

IMMUNIZATION

The Centers for Disease Control and Prevention (CDC) makes annual recommendations for immunizations in adults ([figure 1](#) and [figure 2](#)). (See ["Standard immunizations for nonpregnant adults"](#).)

- **Influenza vaccine** – We recommend annual influenza vaccination for all adults. (See ["Seasonal influenza vaccination in adults"](#).)
- **Tetanus, diphtheria, and acellular pertussis vaccination (Td/Tdap)** – The CDC recommends a single dose of Tdap in place of Td for all adults aged 19 years and older who have not received Tdap previously [52]. Thereafter, all adults should receive a booster every 10 years with either Tdap or Td and should receive a Tdap with each pregnancy. (See ["Tetanus-diphtheria toxoid vaccination in adults"](#) and ["Immunizations during pregnancy"](#).)

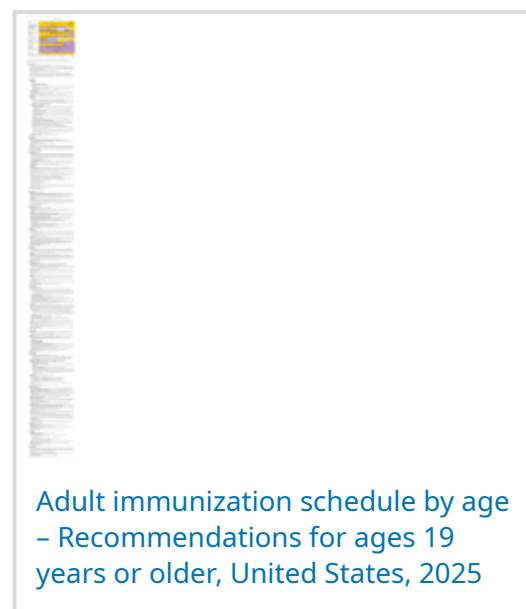


Figure 1 - larger image below

- **Varicella vaccine** – We recommend routine varicella vaccination for healthy persons >13 years of age without evidence of immunity. (See "[Vaccination for the prevention of chickenpox \(primary varicella infection\)](#)", section on 'Adolescents and adults'.)
- **Human papillomavirus (HPV) vaccines** – We recommend routine vaccination with the HPV vaccine for all individuals through age 26 years. (See "[Human papillomavirus vaccination](#)".)
- **Zoster vaccine** – We suggest zoster vaccination in most individuals who are 50 years of age or older. (See "[Vaccination for the prevention of shingles \(herpes zoster\) in adults](#)".)
- **Pneumococcal vaccines** – We recommend pneumococcal vaccination for all adults ≥65 years old, as well as adults 19 and above who have a condition that increases the risk of pneumococcal disease or severe complications from pneumococcal infection ([table 6](#)). We also suggest pneumococcal vaccination for all adults between 50 and 64 years of age. (See "[Pneumococcal vaccination in adults](#)".)
- **Meningococcal vaccines** – Meningococcal vaccination is recommended in the United States based on age and/or risk factors. (See "[Meningococcal vaccination in children and adults](#)", section on 'Indications and schedules in the United States'.)
- **Hepatitis B virus (HBV) vaccination** – HBV vaccination is recommended for all adults <60 years of age, as well as for those ≥60 who are seeking protection against HBV or are at increased risk for acquiring HBV infection ([table 7](#)). (See "[Hepatitis B virus immunization in adults](#)", section on 'Approach to vaccination'.)
- **Coronavirus disease 2019 (COVID-19)** – We recommend that all adults receive at least one dose of an updated (2024 to 2025 formula) COVID-19 vaccine. Immunocompetent individuals ≥65 years of age should receive a second 2024 to 2025 formula COVID-19 dose approximately six months after the first. Recommendations for individuals with immunocompromising conditions ([table 8](#)) vary depending on their vaccination history. (See "[COVID-19: Vaccines](#)", section on 'Approach to vaccination in the United States'.)
- **Respiratory syncytial virus (RSV)** – Several RSV vaccines are available. Recommendations on the use of these vaccines in older adults have been put forth by the [CDC \[53\]](#). Vaccine formulations and the use of RSV vaccines in adults and pregnant persons are discussed

Indications for pneumococcal vaccination in adults in the United States
All adults ≥50 years of age Adults 19 to 49 years of age with any of the following: • Preexisting medical conditions: • Alcohol use disorder • Chronic heart disease^a • Chronic lung disease^a • Chronic liver disease • Diabetes mellitus • Sickle cell disease or other hemoglobinopathies • Current cigarette smoking • Increased risk of meningitis: • Cerebrospinal fluid leak • Cochlear implant • Immunocompromising conditions and other conditions associated with altered immunocompetence^b: • Congenital or acquired immunodeficiency^c • Generalized active malignancy • HIV infection^d • Steroid immunosuppression^e • Hodgkin disease • Leukemia • Lymphoma • Multiple myeloma • Solid organ transplant • Chronic kidney disease^f and nephrotic syndrome • Functional or anatomic asplenia • History of invasive pneumococcal disease^g
Indications for pneumococcal vaccination in adults in the United States

Table 6 - larger image below

elsewhere. (See ["Respiratory syncytial virus infection in adults"](#) and ["Immunizations during pregnancy"](#), section on 'Respiratory syncytial virus'.)

Additional immunizations may be necessary for patients in particular risk groups (eg, health care workers, asplenia, individuals with cancer), and there are considerations for appropriate vaccinations during pregnancy. Please see the appropriate topics for details. (See ["Immunizations for health care providers"](#) and ["Immunizations in adults with cancer"](#) and ["Prevention of infection in patients with impaired splenic function"](#), section on 'Vaccinations' and ["Immunizations during pregnancy"](#).)

SEXUALLY TRANSMITTED AND BLOODBORNE INFECTIONS

Prevention and screening for sexually transmitted infections in asymptomatic patients and pregnant individuals are discussed in detail elsewhere. (See ["Prevention of sexually transmitted infections"](#) and ["Screening for sexually transmitted infections"](#) and ["Prenatal care: Initial assessment"](#), section on 'Infection'.)

Additional screening may be necessary for patients in particular risk groups (eg, commercial sex workers ([table 9](#)), men who have sex with men ([table 10](#)), persons in correctional facilities, persons diagnosed with another sexually transmitted infection). (See ["Screening for sexually transmitted infections"](#).)

- **Chlamydia** – We suggest chlamydia screening for all sexually active women <25 years and sexually active women ≥25 years with risk factors (eg, history of prior chlamydial or other sexually transmitted infection, new or multiple sex partners, sex partner with concurrent partners, sex partner with a sexually transmitted infection, exchanging sex for drugs or money). (See ["Screening for sexually transmitted infections"](#), section on 'Female patients'.)

We also suggest screening men in high-risk populations (men attending clinics for sexually transmitted infections, men who have sex with men, or men in correctional facilities). (See ["Screening for sexually transmitted infections"](#), section on 'Male patients'.)

- **Gonorrhea** – Routine screening should be offered to sexually active patients at high risk of infection, including (see ["Clinical manifestations and diagnosis of *Neisseria gonorrhoeae* infection in adults and adolescents"](#), section on 'Testing asymptomatic individuals'):
 - HIV-infected persons
 - Sexually active women <25 years old
 - Individuals with new or many sexual partners
 - Men who have sex with men
 - Sexually active individuals living in areas of high *Neisseria gonorrhoeae* prevalence
 - Individuals with a history of other sexually transmitted infection(s)

- Women ≤35 years old and men ≤30 years old entering correctional facilities, at intake
- **Hepatitis B** – In general, we suggest one-time universal screening for hepatitis B virus (HBV) for adults >18 years of age. Unvaccinated persons at increased risk may need repeated screening ([table 11](#)). More detailed information on hepatitis B screening is presented separately. (See "[Hepatitis B virus: Screening and diagnosis in adults](#)", section on '[Approach to screening and testing](#)').)
- **Hepatitis C** – We recommend a one-time screening for all United States adults ages between 18 to 79 years. Rescreening is appropriate for those with risk factors that may result in continued exposure to hepatitis C; this is reviewed elsewhere. (See "[Screening and diagnosis of chronic hepatitis C virus infection](#)" and "[Screening and diagnosis of chronic hepatitis C virus infection](#)", section on '[Those with ongoing risk](#)').)
- **HIV infection** – We recommend screening adults for HIV infection until age 75. (See "[Screening and diagnostic testing for HIV infection in adults](#)".)
- **Syphilis** – We recommend screening for syphilis in patients considered high risk. These include men who have sex with men, individuals in prison, persons with multiple sexual partners, patients with another sexually transmitted infection, and sexually active HIV-infected patients. (See "[Syphilis: Screening and diagnostic testing](#)", section on '[Whom to test](#)').)

Groups at increased risk for hepatitis B virus infection
Individuals at risk for HBV due to vertical transmission (ie, mother-to-child transmission) - Individuals born in regions with high (>8%) or intermediate (2-8%) prevalence rates for HBV, including immigrants and adopted children^a - Infants born to pregnant persons who are HBsAg positive^a - US-born persons not vaccinated as infants whose parents were born in regions with high HBV endemicity (>8%)^a
Individuals at risk due to horizontal transmission (ie, percutaneous or mucosal exposure to blood or body fluids contaminated with blood)^a - Household contacts of HBsAg-positive persons - Needle sharing or sexual contacts of HBsAg-positive persons - Individuals who have ever injected drugs - Individuals with multiple sexual partners and/or history of sexually transmitted infections - Men who have sex with men - Inmates of correctional facilities or other detention settings - Individuals with HIV infection^b - Individuals with current or past HCV infection^b - Individuals with end-stage kidney disease on maintenance renal dialysis
Other individuals - Individuals with elevated alanine aminotransferase or aspartate aminotransferase levels of unknown etiology - Individuals who request HBV testing

Groups at increased risk for hepatitis B virus infection

Table 11 - larger image below

PSYCHOSOCIAL HEALTH CONCERNS

Depression — We suggest screening adolescents and adults (ages 12 and older) for depression, including older adults. Screening is most effective when personnel or systems are in place to ensure appropriate follow-up and management for those who screen positive. (See "[Screening for depression in adults](#)", section on '[Who to screen](#)' and "[Screening for depression in adults](#)", section on '[Practice support](#)'.)

We suggest using a two-question screen or the Patient Health Questionnaire-2 (PHQ-2) because of ease of administration and reasonable sensitivity ([table 12](#)). A positive screen should be followed by a thorough clinical assessment, facilitated with the PHQ-9 ([table 13](#))

or a similar instrument, to diagnose depression. (See ["Screening for depression in adults", section on 'Two-step approach to screening'.](#))

Limited evidence guides the optimal frequency of screening for depression. To facilitate ease of implementation, we support screening at the time of a routine health visit. (See ["Screening for depression in adults", section on 'Practical implementation, follow-up, and practice support'.](#))

Anxiety — We suggest screening adolescents and adults ages 13 to 64 for anxiety disorders. Screening should use a validated instrument, such as the Generalized Anxiety Disorder-7 (GAD-7) scale. Although the benefits and harms of screening adults ages 65 and older are likely similar to those for younger individuals, insufficient evidence exists to support screening in this population. Because few data exist to clarify the optimal timing or frequency of screening, we support screening during a routine health visit. (See ["Generalized anxiety disorder in adults: Epidemiology, pathogenesis, clinical manifestations, course, assessment, and diagnosis", section on 'Screening'.](#))

Substance use-related problems

- **Alcohol** – We recommend that all adult primary care patients be screened for unhealthy alcohol use. In general, annual screening is suggested. We suggest the single-item alcohol screening question for most primary care practices. Single-item screen starts with the question, "Do you sometimes drink beer, wine, or other alcoholic beverages?" If yes, then the clinician can follow up with, "How many times in the past year have you had five (four for women) or more drinks in a day?"

The Alcohol Use Disorders Identification Test – Consumption (AUDIT-C) ([table 14](#)) is an alternative for offices with more resources (eg, staff, time, ability to automate administration and scoring). (See ["Screening for unhealthy use of alcohol and other drugs in primary care".](#))

- **Other drug use** – We also suggest screening patients for unhealthy use of drugs, including illegal drugs and prescription medications that are not used for medical purposes [54]. (See ["Screening for unhealthy use of alcohol and other drugs in primary care".](#))

Intimate partner violence — Given the high prevalence of intimate partner violence and the lack of harm and potential benefits of screening, we suggest routine screening for all patients on all visits to primary care clinicians, obstetrician-gynecologists, and emergency departments, as well as on hospital admission. (See ["Intimate partner violence: Diagnosis and screening".](#))

OSTEOPOROSIS

We suggest osteoporosis screening in the following groups (see ["Screening for osteoporosis in men aged ≥50 years and postmenopausal women"](#) and ["Fracture risk assessment"](#)):

- Postmenopausal women <65 years with risk factors for osteoporosis ([table 15](#)).
- Men with clinical manifestation of low bone mass, history of low-trauma fracture, risk factors for fracture (such as androgen deprivation therapy for prostate cancer, hypogonadism, primary hyperparathyroidism, or intestinal disorders).
- All women ≥65 years. (See ["Geriatric health maintenance"](#), section on 'Osteoporosis'.)

Clinical risk factors for fracture
Advanced age (eg, ≥65 years for women)*
Previous fracture during adulthood
High fall risk
Impaired mobility
Chronic glucocorticoid therapy (eg, ≥3 months)
Parental history of hip fracture
Low body weight (<58 kg [127 lb])
Current cigarette smoking
Excess alcohol intake
Height loss (>3.8 cm [1.5 in]) and/or kyphosis
Secondary osteoporosis (eg, rheumatoid arthritis, hyperthyroidism, primary hyperparathyroidism, hypogonadal states, gastrointestinal diseases and malabsorptive states)

Clinical risk factors for fracture

Table 15 - larger image below

There are several modalities to evaluate bone density. Dual-energy x-ray absorptiometry (DXA) is most commonly used. (See ["Overview of dual-energy x-ray absorptiometry"](#).)

VASCULAR DISEASE

- **Abdominal aortic aneurysm (AAA)** – Screening for AAA is not recommended for adults <65 years. (See ["Screening for abdominal aortic aneurysm"](#).)
 - We recommend one-time ultrasound screening for AAA in men ages 65 to 75 years who are current or former smokers.
 - We also suggest one-time ultrasound screening for AAA in men ages 65 to 75 who have never smoked but who have a first-degree relative who required repair of an AAA or died from a ruptured AAA.
- **Carotid artery stenosis** – We do not recommend screening (either by auscultation for carotid bruit or by carotid ultrasound) asymptomatic patients for carotid artery stenosis. (See ["Screening for asymptomatic carotid artery stenosis"](#).)
- **Peripheral artery disease** – We do not suggest screening for peripheral artery disease. (See ["Asymptomatic peripheral artery disease"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th

grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see ["Patient education: Routine physical for adults \(The Basics\)"](#) and ["Patient education: Preventive health care for ages 18 to 39 \(The Basics\)"](#) and ["Patient education: Preventive health care for ages 40 to 64 \(The Basics\)"](#) and ["Patient education: Lowering your risk of heart disease \(The Basics\)"](#) and ["Patient education: Cancer screening \(The Basics\)"](#) and ["Patient education: Vaccines for adults \(The Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- **Prioritizing preventive care** – Appropriate screening and prevention interventions need to be prioritized both for effectiveness in modifying target outcomes and for risk of the condition in the individual patient. (See ["Prioritizing preventive services"](#) above.)
- **Periodic health maintenance visits** – We suggest periodic health maintenance visits every three years for adult patients ≤49 years without chronic conditions and annually for adults ≥50 years. Preventive counseling and services are important components of the periodic health check-up. (See ["Periodic "check-up" above.](#))
- **Challenges of delivering screening and counseling interventions** – Patients may not receive appropriate screening and prevention interventions if the time for addressing these needs is limited to a "routine physical examination." Creative solutions for delivering preventive care include involving use of patient databases and mechanisms for patient outreach, as well as incorporating preventive care into patient sick visits. (See ["Supportive office systems"](#) above.)
- **Important targets for disease prevention**
 - **Cardiovascular disease (CVD)** – Key features of CVD prevention include risk assessment and treatment of predisposing conditions. (See ["Cardiovascular disease prevention"](#) above.)
 - **Cancer** – Cancer prevention includes counseling to reduce risk factors, most notably smoking, and appropriate screening. (See ["General cancer prevention counseling"](#) above)

and ['Cancer screening'](#) above.)

- **Sexually transmitted and bloodborne infections** – Screening for sexually transmitted infections and bloodborne infections is based on age and patient risk. (See ['Sexually transmitted and bloodborne infections'](#) above.)
- **Psychologic health conditions** – Adults should receive screening for psychologic conditions such as depression, anxiety, and substance use and referrals for treatment if indicated. (See ['Psychosocial health concerns'](#) above.)
- **Immunization** – Immunization is an effective method of disease prevention. The Centers for Disease Control and Prevention (CDC) makes annual recommendations for immunizations in adults ([figure 1](#) and [figure 2](#)). (See ['Immunization'](#) above.)

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GRAPHICS

Table 1: Definition of hypertension based on blood pressure measurement strategy

SBP/DBP	Clinic	SMBP	Daytime ABPM	Nighttime ABPM	24-hour ABPM
ACC/AHA guidelines 2025 ^[1]	≥130/80	≥130/80	≥130/80	≥110/65	≥125/75
ESC guidelines 2024 ^[2]	≥140/90	≥135/85	≥135/85	≥120/70	≥130/80

ABPM: ambulatory blood pressure monitoring; ACC/AHA: American College of Cardiology/American Heart Association; DBP: diastolic blood pressure; ESC: European Society of Cardiology; SBP: systolic blood pressure; SMBP: self-measured blood pressure.

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Table 2: Factors associated with increased risk of diabetes

▪ Overweight or obesity (BMI ≥25 kg/m² or ≥23 kg/m² in Asian Americans) ▪ Gestational diabetes ▪ First-degree relative with diabetes ▪ Polycystic ovary syndrome ▪ Hypertension (≥130/80 mmHg or on therapy for hypertension) ▪ High-risk race/ethnicity (eg, African American, Latino, Native American, Asian American, Pacific Islander) ▪ Established cardiovascular disease (eg, MI, heart failure) ▪ HDL cholesterol level <35 mg/dL (<0.90 mmol/L) and/or a triglyceride level >250 mg/dL (>2.82 mmol/L) ▪ Metabolic syndrome ▪ Physical inactivity ▪ Other clinical conditions associated with insulin resistance (eg, acanthosis nigricans) ▪ HIV disease ▪ Exposure to high-risk medications (eg, glucocorticoids, second-generation antipsychotics, HIV antiretrovirals, immune checkpoint inhibitors, thiazide diuretics, high-potency statins) ▪ History of pancreatitis
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BMI: body mass index; HDL: high-density lipoprotein; HIV: human immunodeficiency virus; MI: myocardial infarction.

References:

1. American Diabetes Association Professional Practice Committee. *Diagnosis and classification of diabetes: Standards of care in diabetes—2024*. *Diabetes Care* 2024; 47:S20.
2. Crawford AL, Laiteerapong N. *Type 2 diabetes mellitus*. *Ann Intern Med* 2024; 177:ITC81.

Table 3: Risk and protective factors for developing breast cancer

	Risk group		
	Low risk	High risk	Relative risk
Risk factors			
Deleterious <i>BRCA1/BRCA2</i> genes	Negative	Positive	3.0 to 7.0
Mother or sister with breast cancer	No	Yes	2.6
Age	30 to 34	70 to 74	18.0
Age at menarche	>14	<12	1.5
Age at first birth	<20	>30	1.9 to 3.5
Age at menopause	<45	>55	2.0
Use of contraceptive pills	Never	Past/current use	1.07 to 1.2
Hormone replacement therapy (estrogen + progestin)	Never	Current	1.2
Alcohol	None	2 to 5 drinks/day	1.4
Breast density on mammography (percents)	0	≥75	1.8 to 6.0
Bone density	Lowest quartile	Highest quartile	2.7 to 3.5
History of a benign breast biopsy	No	Yes	1.7
History of atypical hyperplasia on biopsy	No	Yes	3.7
Protective factors			
Breast feeding (months)	≥16	0	0.73
Parity	≥5	0	0.71
Recreational exercise	Yes	No	0.70
Postmenopause body mass index (kg/m ²)	<22.9	>30.7	0.63
Oophorectomy before age 35 years	Yes	No	0.3
Aspirin	≥Once/week for ≥6 months	Nonusers	0.79

Adapted from: Clemons M, Goss P. Estrogen and the risk of breast cancer. *N Engl J Med* 2001; 344:276.

Table 4: Risk assessment criteria for inherited breast-ovarian cancer syndrome, combining several guidelines

Non-Jewish families
Any of the following:
1 case of breast cancer ≤ 40 years old in a first- or second-degree relative
1 first- or second-degree relative with breast and ovarian cancer, at any age
2 or more cases of breast cancer in first- or second-degree relatives if one is diagnosed at ≤ 50 years old or is bilateral cancer
1 first- or second-degree relative with breast cancer at ≤ 50 years old, or bilateral breast cancer, and 1 first- or second-degree relative with ovarian cancer
3 cases of breast and ovarian cancer (at least 1 case of ovarian cancer) in first- or second-degree relatives
2 cases of ovarian cancer in first- or second-degree relatives
1 case of male breast cancer in a first- or second-degree relative if another first- or second-degree relative has (male or female) breast or ovarian cancer
Jewish families
Any of the following:
1 or more cases of breast cancer ≤ 50 years old in a first- or second-degree relative
1 or more cases of ovarian cancer at any age in a first- or second-degree relative
1 or more first- or second-degree relatives with breast cancer at any age, if another first- or second-degree relative has breast and/or ovarian cancer at any age
1 or more cases of male breast cancer in a first- or second-degree relative

Adapted from: Hampel H, Sweet K, Westman JA, et al. Referral for cancer genetics consultation: a review and compilation of risk assessment criteria. J Med Genet 2004; 41:81.

Table 5: Society and expert recommendations for routine mammographic screening in women at average risk

Group (date)	Frequency of screening (years)	Initiation of screening for women at average risk		
		40 to 49 years of age	50 to 69 years of age	≥70 years of age
Government-sponsored groups				
United States Preventive Services Task Force (2024) ^[1]	2	Yes, start age 40	Yes	Yes, to age 74
Canadian Task Force on Preventive Health Care (2018) ^[2]	2 to 3	Recommend against [*]	Yes	Yes, to age 74
National Health Service, United Kingdom (2018) ^[3]	3	No	Yes	Yes, to age 71
Royal Australian College of General Practitioners (2018) ^[4]	2	No	Yes	Yes, to age 74
Medical societies				
American College of Obstetricians and Gynecologists (2017) ^[5]	1 to 2 [*]	Yes	Yes	Yes, to at least age 75 [¶]
American College of Physicians (2025) ^[6]	2	Individualize [*]	Yes	Yes, to age 74 [¶]
American Cancer Society (2015) ^[7]	1 year age 45 to 54	Individualize [*] through age 44	Yes	Yes ^Δ
	1 to 2 years age ≥55	Yes, start age 45		
American College of Radiology and Society of Breast Imaging (2021) ^[8]	1	Yes	Yes	Yes [◇]
Coalitions				
National Comprehensive Cancer Network (2018) ^[9]	1	Yes	Yes	Yes

^{*} Women should be counseled about the harms and benefits of mammography; individualized decisions should include shared decision-making based on risks, benefits, patient values, and preferences.

[¶] Decision to discontinue screening mammography should be based on a shared decision-making process informed by the woman's health status and longevity.

^Δ If in good health and life expectancy >10 years.

◇ Individualize to current health and life expectancy; if a woman is in reasonably good health and would be a candidate for treatment, then should continue screening.

References:

1. *Final recommendation statement: Breast cancer: Screening.* US Preventive Services Task Force. <https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/breast-cancer-screening#bcei-recommendation-title-area> (Accessed on July 15, 2026).
 2. Canadian Task Force on Preventive Health Care, Klarenbach S, Sims-Jones N, Lewin G, et al. Recommendations on screening for breast cancer in women aged 40-74 years who are not at increased risk for breast cancer. *CMAJ* 2018; 190:E1441.
 3. National Health Service. When it's offered: Breast cancer screening. <https://www.nhs.uk/conditions/breast-cancer-screening/when-its-offered/> (Accessed on July 15, 2026).
 4. The Royal Australian College of General Practitioners. *Guidelines for preventive activities in general practice*, 9th ed, East Melbourne, RACGP 2018.
 5. Committee on Practice Bulletins—Gynecology. Practice bulletin no. 179: Breast cancer risk assessment and screening in average-risk women. *Obstet Gynecol* 2017; 130:e1-16.
 6. Qaseem A, Harrod CS, Balk EML, et al. Screening for breast cancer in asymptomatic, average-risk adult females: A guidance statement from the American College of Physicians (version 2). *Ann Intern Med* 2026; 179:842.
 7. Oeffinger KC, Fontham ETH, Etzioni R, et al. Breast cancer screening for women at average risk: 2015 guideline update from the American Cancer Society. *JAMA* 2015; 314:1599.
 8. Monticciolo DL, Malak SF, Friedewald SM, et al. Breast cancer screening recommendations inclusive of all women at average risk: Update from the ACR and Society of Breast Imaging. *J Am Coll Radiol* 2021; 18:1280.
 9. NCCN clinical practice guidelines in oncology: Breast cancer, version 2. National Comprehensive Cancer Network 2018.
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Figure 1: Adult immunization schedule by age – Recommendations for ages 19 years or older, United States, 2025

	Age group (years)						
Vaccine	19 through 26 years		27 through 49 years		50 through 64 years	≥65 years	
COVID-19*	1 or more doses of 2024-2025 vaccine (refer to footnotes)					2 or more doses of 2024-2025 vaccine (refer to footnotes)	
Influenza inactivated (IIV3, ccIIV3)¶ Influenza recombinant (RIV3)¶	1 dose annually					1 dose annually (HD-IIV3, RIV3, or aIIV3 preferred)	
Influenza inactivated (aIIV3; HD-IIV3)¶ Influenza recombinant (RIV3)¶	Solid organ transplant (refer to footnotes)						
Influenza live, attenuated (LAIV3) ¶	1 dose annually						
Respiratory syncytial virus (RSV)Δ	Seasonal administration during pregnancy. (Refer to footnotes.)				60 through 74 years (refer to footnotes)		≥75 years
Tetanus, diphtheria, pertussis (Tdap or Td)◊	1 dose Tdap each pregnancy; 1 dose Td/Tdap for wound management (refer to footnotes)						
	1 dose Tdap, then Td or Tdap booster every 10 years						
Measles, mumps, rubella (MMR)§	1 or 2 doses depending on indication (if born in 1957 or later)					For health care personnel (refer to footnotes)	
Varicella (VAR)⌘	2 doses (if born in 1980 or later)			2 doses			
Zoster recombinant (RZV)‡	2 doses for immunocompromising conditions (refer to footnotes)			2 doses			
Human papillomavirus (HPV)†	2 or 3 doses depending on age at initial vaccination or condition	27 through 45 years					
Pneumococcal (PCV15, PCV20, PCV21, PPSV23)**				Refer to footnotes			
				Refer to footnotes			
Hepatitis A (HepA) ¶¶	2, 3, or 4 doses depending on vaccine						
Hepatitis B (HepB)ΔΔ	2, 3, or 4 doses depending on vaccine or				condition		
Meningococcal A, C, W, Y (MenACWY)◊◊	1 or 2 doses depending on indication (refer to footnotes for booster recommendations)						
Meningococcal B (MenB)◊◊	2 or 3 doses depending on vaccine and indication (refer to footnotes for booster recommendations)						
	19 through 23 years						
Haemophilus influenzae type b (Hib)§§	1 or 3 doses depending on indication						
Mpox⌘⌘	2 doses						
Inactivated poliovirus (IPV)⌘⌘	Complete 3-dose series if incompletely vaccinated. Self-report of previous doses acceptable. (Refer to footnotes.)						

Recommended vaccination for adults who meet age requirement, lack documentation of vaccination, or lack evidence of immunity
 Recommended vaccination for adults with an additional risk factor or another indication
 Recommended vaccination based on shared clinical decision-making
 No guidance/ not applicable

Administer recommended vaccines if vaccination history is incomplete or unknown. Do not restart or add doses to vaccine series if there are extended intervals between doses. The use of trade names is for identification purposes only and does not imply endorsement by the ACIP or CDC.

NOTES

For vaccine recommendations for persons 18 years of age or younger, refer to the [Recommended Child and Adolescent Immunization Schedule](#).

Additional information

- For calculating intervals between doses, 4 weeks = 28 days. Intervals of ≥ 4 months are determined by calendar months.
- Within a number range (eg, 12-18), a dash (–) should be read as "through."
- Vaccine doses administered ≤ 4 days before the minimum age or interval are considered valid. Doses of any vaccine administered ≥ 5 days earlier than the minimum age or minimum interval should not be counted as valid and should be repeated. **The repeat dose should be spaced after the invalid dose by the recommended minimum interval.** For further details, refer to Table 3-2, Recommended and minimum ages and intervals between vaccine doses, in [Timing and Spacing of Immunobiologics](#).
- Information on travel vaccination requirements and recommendations is available at cdc.gov/travel/.
- For vaccination of persons with immunodeficiencies, refer to Table 8-1, Vaccination of persons with primary and secondary immunodeficiencies, in [Altered Immunocompetence](#).
- For information about vaccination in the setting of a vaccine-preventable disease outbreak, contact your state or local health department.
- The National Vaccine Injury Compensation Program (VICP) is a no-fault alternative to the traditional legal system for resolving vaccine injury claims. All vaccines included in the adult immunization schedule except PPSV23, RSV, RZV, mpox, and COVID-19 vaccines are covered by the National Vaccine Injury Compensation Program (VICP). Mpox and COVID-19 vaccines are covered by the Countermeasures Injury Compensation Program (CICP). For more information, refer to www.hrsa.gov/vaccinecompensation or www.hrsa.gov/cicp.

* COVID-19 vaccination

- **Routine vaccination:**
 - **Age 19-64 years.**
 - **Unvaccinated:**
 - 1 dose 2024-25 Moderna or Pfizer-BioNTech.
 - 2 doses 2024-25 Novavax at 0, 3-8 weeks.
 - **Previously vaccinated before 2024-25 vaccine with:**
 - **1 or more doses Moderna or Pfizer-BioNTech:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech at least 8 weeks after the most recent dose.
 - **1 dose Novavax:** 1 dose 2024-25 Novavax 3-8 weeks after most recent dose. If more than 8 weeks after the most recent dose, administer 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech.
 - **2 or more doses Novavax:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech at least 8 weeks after the most recent dose.
 - **1 or more doses Janssen:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech.
 - **Age 65 years and older.**
 - **Unvaccinated:** Follow recommendations above for unvaccinated persons ages 19-64 years **and** administer dose 2 of 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months).

- **Previously vaccinated before 2024-25 vaccine:** Follow recommendations above for previously vaccinated persons ages 19-64 years **and** administer dose 2 of 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months).
- **Special situations:**
 - **Persons who are moderately or severely immunocompromised. Use vaccine from the same manufacturer for all doses in the initial vaccination series.**
 - **Unvaccinated:**
 - 4 doses (3-dose initial series 2024-25 Moderna at 0, 4 weeks, and at least 4 weeks after dose 2, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses.^{††}
 - 4 doses (3-dose initial series 2024-25 Pfizer-BioNTech at 0, 3 weeks, and at least 4 weeks after dose 2, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses.^{††}
 - 3 doses (2-dose initial series 2024-25 Novavax at 0, 3 weeks, followed by 1 dose Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses.^{††}
 - **Incomplete initial vaccination series before 2024-25 vaccine:**
 - **Previous vaccination with Moderna**
 - **1 dose Moderna:** Complete initial series with 2 doses 2024-25 Moderna at least 4 weeks apart (administer dose 1 4 weeks after most recent dose), followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses.^{††}
 - **2 doses Moderna:** Complete initial series with 1 dose 2024-25 Moderna at least 4 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses.^{††}
 - **Previous vaccination with Pfizer-BioNTech**
 - **1 dose Pfizer-BioNTech:** Complete initial series with 2 doses 2024-25 Pfizer-BioNTech at least 4 weeks apart (administer dose 1 3 weeks after most recent dose), followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses.^{††}
 - **2 doses Pfizer-BioNTech:** Complete initial series with 1 dose 2024-25 Pfizer-BioNTech at least 4 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses.^{††}
 - **Previous vaccination with Novavax**
 - **1 dose Novavax:** Complete initial series with 1 dose 2024-25 Novavax at least 3 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses.^{††}
 - **Completed the initial vaccination series before 2024-25 vaccine with:**
 - **3 or more doses Moderna or 3 or more doses Pfizer-BioNTech:** 2 doses 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months apart (minimum interval 2 months). Administer dose 1 at least 8 weeks after the most recent dose. May administer additional doses.^{††}
 - **2 or more doses Novavax:** 2 doses 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months apart (minimum interval 2 months). Administer dose 1 at least 8 weeks after the most recent dose. May administer additional doses.^{††}

- †† **Additional doses of 2024-25 COVID-19 vaccine for moderately or severely immunocompromised:**
 - Based on shared clinical decision-making and administered at least 2 months after the most recent dose (refer to Table 2 at www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#table-02). For description of moderate and severe immunocompromising conditions and treatment, refer to www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#immunocompromising-conditions-treatment.
 - Unvaccinated persons have never received any COVID-19 vaccine doses. There is no preferential recommendation for the use of one COVID-19 vaccine over another when more than one recommended age-appropriate vaccine is available. Administer an age-appropriate COVID-19 vaccine product for each dose.
 - For more information about the interchangeability of COVID-19 vaccines, refer to <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#Interchangeability>.
 - Current COVID-19 schedule and dose formulation available at www.cdc.gov/covidschedule.
- **Contraindications and precautions:**
 - For contraindications and precautions to COVID-19 vaccination, refer to [COVID-19 Appendix](#).

¶ Influenza vaccination

- **Routine vaccination:**
 - **Age 19 years or older:** 1 dose any influenza vaccine appropriate for age and health status annually.
 - **Solid organ transplant recipients aged 19 through 64 years receiving immunosuppressive medications:** HD-IIV3 and aIIV3 are acceptable options. No preference over other age-appropriate IIV3 or RIV3.
 - **Age 65 years or older:** Any one of HD-IIV3, RIV3, or aIIV3 is preferred. If none of these three vaccines are available, then any other age-appropriate influenza vaccine should be used.
 - For the 2024-25 season, refer to www.cdc.gov/mmwr/volumes/73/rr/rr7305a1.htm.
 - For the 2025-26 season, refer to the 2025-26 ACIP influenza vaccine recommendations.
- **Special situations:**
 - **Close contacts (eg, caregivers, health care workers) of severely immunosuppressed persons who require a protected environment:** Should not receive LAIV3. If LAIV3 is given, they should avoid contact with/caring for such immunosuppressed persons for 7 days after vaccination.
 - **NOTE:** Persons with an egg allergy can receive any influenza vaccine (egg-based and non-egg-based) appropriate for age and health status.
- **Contraindications and precautions:**
 - For contraindications and precautions to influenza vaccination, refer to [IIV3 Appendix](#), [LAIV3 Appendix](#), [ccIIV3 Appendix](#), and [RIV3 Appendix](#).

Δ Respiratory syncytial virus vaccination

- **Routine vaccination:**
 - **Pregnant persons of any age.**
 - **Pregnant at 32 weeks 0 days through 36 weeks and 6 days gestation from September through January in most of the continental United States** (NOTE: Providers in jurisdictions with RSV seasonality that differs from most of the continental United States [eg, Alaska, jurisdiction with tropical climate] should follow guidance from public health authorities on timing of administration. Refer to the 2025 Child and Adolescent

Immunization Schedule for considerations regarding nirsevimab administration to infants): 1 dose Abrysvo. Administer RSV vaccine regardless of previous RSV infection.

- Either maternal RSV vaccination with Abrysvo or infant immunization with nirsevimab (RSV monoclonal antibody) is recommended to prevent severe respiratory syncytial virus disease in infants.
- **All other pregnant persons:** RSV vaccine not recommended.
- **Subsequent pregnancies:** Additional doses not recommended. No data are available to inform whether additional doses are needed in subsequent pregnancies. Infants born to pregnant persons who received RSV vaccine during a previous pregnancy should receive nirsevimab.

- **Age 75 years or older.**

- **Unvaccinated:** 1 dose (Arexvy or Abrysvo or mResvia). Additional doses not recommended.
- **Previously vaccinated:** Additional doses not recommended. No data are available to inform whether additional doses are needed.

- **Special situations:**

- **Age 60-74 years.**

- **Unvaccinated and at increased risk of severe RSV disease^{***}:** 1 dose (Arexvy or Abrysvo or mResvia). Additional doses not recommended.
- **Previously vaccinated:** Additional doses not recommended. No data are available to inform whether additional doses are needed.

- Persons 60 years and older can get RSV vaccine at any time but best to administer in late summer and early fall before RSV spreads in communities—ideally August through October in most of continental United States. For further guidance, refer to www.cdc.gov/mmwr/volumes/73/wr/mm7332e1.htm.

- ***** People can self-attest to the presence of a risk factor. The following medical and other conditions increase the risk of severe RSV disease:**

- Chronic cardiovascular disease (eg, heart failure, coronary artery disease, congenital heart disease). Excludes isolated hypertension.
- Chronic lung or respiratory disease (eg, chronic obstructive pulmonary disease, emphysema, asthma, interstitial lung disease, cystic fibrosis).
- End-stage kidney disease, dependence on hemodialysis, or other kidney replacement therapy.
- Diabetes mellitus complicated by chronic kidney disease, neuropathy, retinopathy, or other end-organ damage.
- Diabetes mellitus requiring treatment with insulin or sodium-glucose cotransporter 2 (SGLT2) inhibitor.
- Neurologic or neuromuscular conditions causing impaired airway clearance or respiratory muscle weakness (eg, post-stroke dysphagia, amyotrophic lateral sclerosis, muscular dystrophy). Excludes history of stroke without impaired airway clearance.
- Chronic liver disease (eg, cirrhosis).
- Chronic hematologic conditions (eg, sickle cell disease, thalassemia).
- Severe obesity (body mass index ≥ 40 kg/m²).
- Moderate or severe immune compromise.
- Residence in a nursing home.
- Other chronic medical conditions or risk factors that a health care provider determines would increase the risk of severe disease due to viral respiratory infection (eg, frailty, concern for presence of undiagnosed chronic medical conditions, residence in a remote or rural community where escalation of medical care is challenging).

- **Contraindications and precautions:**

- For contraindications and precautions to RSV vaccine, refer to [RSV Appendix](#).
-

◇ Tetanus, diphtheria, and pertussis vaccination

▪ Routine vaccination:

- **Completed primary series and received at least 1 dose Tdap at age 10 years or older:** Td or Tdap every 10 years thereafter.
- **Completed primary series and did NOT receive Tdap at age 10 years or older:** 1 dose Tdap, then Td or Tdap every 10 years thereafter.
- **Unvaccinated or incomplete primary vaccination series for tetanus, diphtheria, or pertussis:** Administer remaining doses (1, 2, or 3 doses) to complete 3-dose primary series. 1 dose Tdap followed by 1 dose Td or Tdap at least 4 weeks later, and a third dose of Td or Tdap 6-12 months later (Tdap is preferred as first dose and can be substituted for any Td dose), then Td or Tdap every 10 years thereafter.

▪ Special situations:

- **Pregnancy:** 1 dose Tdap during each pregnancy, preferably in early part of gestational weeks 27-36.
- **Wound management:** Persons with 3 or more doses of tetanus-toxoid-containing vaccine: For clean and minor wounds, administer Tdap or Td if more than 10 years since last dose of tetanus-toxoid-containing vaccine; for all other wounds, administer Tdap or Td if more than 5 years since last dose of tetanus-toxoid-containing vaccine. Tdap is preferred for persons who have not previously received Tdap or whose Tdap history is unknown. If a tetanus-toxoid-containing vaccine is indicated for a pregnant person, use Tdap. For detailed information, refer to www.cdc.gov/mmwr/volumes/69/wr/mm6903a5.htm.

▪ Contraindications and precautions:

- For contraindications and precautions to tetanus, diphtheria, and acellular pertussis (Tdap), refer to [Tdap Appendix](#).

§ Measles, mumps, and rubella vaccination

▪ Routine vaccination:

- **No evidence of immunity to measles, mumps, or rubella:** 1 dose.
 - **Evidence of immunity:** Born before 1957 (except for health care personnel, refer below), documentation of receipt of MMR vaccine, laboratory evidence of immunity or disease (diagnosis of disease without laboratory confirmation is not evidence of immunity).

▪ Special situations:

- **Pregnancy with no evidence of immunity to rubella:** MMR contraindicated during pregnancy; after pregnancy (before discharge from health care facility), 1 dose.
- **Nonpregnant persons of childbearing age with no evidence of immunity to rubella:** 1 dose.
- **Human immunodeficiency virus (HIV) infection with CD4 percentages $\geq 15\%$ and CD4 count ≥ 200 cells/mm³ for at least 6 months and no evidence of immunity to measles, mumps, or rubella:** Complete 2-dose series at least 4 weeks apart; MMR contraindicated for HIV infection with CD4 percentage $< 15\%$ or CD4 count < 200 cells/mm³.
- **Severe immunocompromising conditions:** MMR contraindicated.
- **Students in postsecondary educational institutions, international travelers, and household or close, personal contacts of immunocompromised persons with no evidence of immunity to measles, mumps, or rubella:** Complete 2-dose series at least 4 weeks apart if previously did not receive any doses of MMR or 1 dose if previously received 1 dose MMR.
- **In mumps outbreak settings,** for information about additional doses of MMR (including 3rd dose of MMR), refer to www.cdc.gov/mmwr/volumes/67/wr/mm6701a7.htm.

- **Health care personnel:**
 - **Born before 1957 with no evidence of immunity to measles, mumps, or rubella:**
Consider 2-dose series at least 4 weeks apart for protection against measles or mumps or 1 dose for protection against rubella.
 - **Born in 1957 or later with no evidence of immunity to measles, mumps, or rubella:**
Complete 2-dose series at least 4 weeks apart for protection against measles or mumps or at least 1 dose for protection against rubella.
- **Contraindications and precautions:**
 - For contraindications and precautions to measles, mumps, rubella (MMR) vaccine, refer to [MMR Appendix](#).

¥ **Varicella vaccination**

- **Routine vaccination:**
 - **No evidence of immunity to varicella:** 2-dose series 4-8 weeks apart if previously did not receive varicella-containing vaccine (VAR or measles-mumps-rubella-varicella vaccine [MMRV] for children); if previously received 1 dose varicella-containing vaccine, 1 dose at least 4 weeks after first dose.
 - **Evidence of immunity:** United States-born before 1980 (except for pregnant persons and health care personnel [refer below]), documentation of 2 doses varicella-containing vaccine at least 4 weeks apart, diagnosis or verification of history of varicella or herpes zoster by a health care provider, laboratory evidence of immunity or disease.
- **Special situations:**
 - **Pregnancy with no evidence of immunity to varicella:** VAR contraindicated during pregnancy; after pregnancy (before discharge from health care facility), 1 dose if previously received 1 dose varicella-containing vaccine or dose 1 of 2-dose series (dose 2: 4 to 8 weeks later) if previously did not receive any varicella-containing vaccine, regardless of whether United States-born before 1980.
 - **Health care personnel with no evidence of immunity to varicella:** 1 dose if previously received 1 dose varicella-containing vaccine; 2-dose series 4-8 weeks apart if previously did not receive any varicella-containing vaccine, regardless of whether United States-born before 1980.
 - **HIV infection with CD4 percentages $\geq 15\%$ and CD4 count ≥ 200 cells/mm³ with no evidence of immunity:** Vaccination may be considered (2 doses 3 months apart); VAR contraindicated for HIV infection with CD4 percentage $< 15\%$ or CD4 count < 200 cells/mm³.
 - **Severe immunocompromising conditions:** VAR contraindicated.
- **Contraindications and precautions:**
 - For contraindications and precautions to varicella (VAR) vaccine, refer to [VAR Appendix](#).

‡ **Zoster vaccination**

- **Routine vaccination:**
 - **Age 50 years or older** (NOTE: Serologic evidence of prior varicella is not necessary for zoster vaccination. However, if serologic evidence of varicella susceptibility becomes available, providers should follow ACIP guidelines for varicella vaccination first. RZV is not indicated for the prevention of varicella, and there are limited data on the use of RZV in persons without a history of varicella or varicella vaccination): 2-dose series recombinant zoster vaccine (RZV; Shingrix) 2-6 months apart (minimum interval: 4 weeks; repeat dose if administered too soon), regardless of previous herpes zoster or history of zoster vaccine live (ZVL; Zostavax) vaccination.
- **Special situations:**
 - **Pregnancy:** There is currently no ACIP recommendation for RZV use in pregnancy. Consider delaying RZV until after pregnancy.

- **Immunocompromising conditions (including persons with HIV regardless of CD4 count;**
NOTE: If there is no documented history of varicella, varicella vaccination, or herpes zoster, providers should refer to the clinical considerations for use of RZV in immunocompromised adults aged ≥ 19 years and the ACIP varicella vaccine recommendations for further guidance: www.cdc.gov/mmwr/volumes/71/wr/mm7103a2.htm): 2-dose series recombinant zoster vaccine (RZV; Shingrix) 2-6 months apart (minimum interval: 4 weeks; repeat dose if administered too soon). For detailed information, refer to www.cdc.gov/shingles/vaccination/immunocompromised-adults.html.

- **Contraindications and precautions:**

- For contraindications and precautions to zoster recombinant vaccine (RZV), refer to [RZV Appendix](#).

† Human papillomavirus vaccination

- **Routine vaccination:**

- **All persons through age 26 years:** Complete 2- or 3-dose series depending on age at initial vaccination or condition.
 - **Age 9-14 years at initial vaccination and received 1 dose or 2 doses less than 5 months apart:** 1 additional dose.
 - **Age 9-14 years at initial vaccination and received 2 doses at least 5 months apart:** HPV vaccination series complete, no additional dose needed.
 - **Age 15 years or older at initial vaccination:** 3-dose series at 0, 1-2 months, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 12 weeks; dose 1 to dose 3: 5 months; repeat dose if administered too soon).
- No additional dose recommended when any HPV vaccine series of any valency has been completed using the recommended dosing intervals.

- **Shared clinical decision-making:**

- **Adults age 27-45 years:** Based on shared clinical decision-making, complete a 2-dose series (if initiated age 9-14 years) or 3-dose series (if initiated ≥ 15 years).
- For additional information on shared clinical decision-making for HPV; refer to www.cdc.gov/vaccines/hcp/admin/downloads/isd-job-aid-scdm-hpv-shared-clinical-decision-making-hpv.pdf.

- **Special situations:**

- **Age ranges recommended above for routine and catch-up vaccination or shared clinical decision-making also apply in special situations.**
 - **Immunocompromising conditions, including HIV infection:** Complete 3-dose series, even for those who initiate vaccination at age 9 through 14 years.
 - **Pregnancy:** Pregnancy testing is not needed before vaccination. HPV vaccination is not recommended until after pregnancy. No intervention needed if inadvertently vaccinated while pregnant.

- **Further considerations:**

- Some UpToDate contributors support an approach consistent with WHO recommendations: 1 or 2 doses of HPV vaccine for persons age 9-20 years, 2 doses for persons age 21 years or older, and 3 doses for immunocompromised individuals.

- **Contraindications and precautions:**

- For contraindications and precautions to human papillomavirus (HPV) vaccination, refer to [HPV Appendix](#).

** Pneumococcal vaccination

- **Routine vaccination:**

- **Age 50 years or older who have:**

- **Not previously received a dose of PCV13, PCV15, PCV20, or PCV21 or whose previous vaccination history is unknown:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21.
 - If PCV15 is used, administer 1 dose PPSV23 at least 1 year after the PCV15 dose (may use minimum interval of 8 weeks for adults with an immunocompromising condition [NOTE: Immunocompromising conditions include chronic kidney failure, nephrotic syndrome, immunodeficiencies, iatrogenic immunosuppression, generalized malignancy, HIV infection, Hodgkin disease, leukemia, lymphoma, multiple myeloma, solid organ transplant, congenital or acquired asplenia, or sickle cell disease or other hemoglobinopathies], cochlear implant, or cerebrospinal fluid [CSF] leak).
- **Previously received only PCV7:** Follow the recommendation above.
- **Previously received only PCV13:** 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PCV13 dose.
- **Previously received only PPSV23:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PPSV23 dose.
 - If PCV15 is used, no additional PPSV23 doses are recommended.
- **Previously received both PCV13 and PPSV23 but NO PPSV23 was received at age 65 years or older:** 1 dose PCV20 or 1 dose PCV21 at least 5 years after the last pneumococcal vaccine dose.
- **Previously received both PCV13 and PPSV23, AND PPSV23 was received at age 65 years or older:** Based on shared clinical decision-making, 1 dose of PCV20 or 1 dose of PCV21 at least 5 years after the last pneumococcal vaccine dose.
- For guidance on determining which pneumococcal vaccines a patient needs and when, please refer to the mobile app which can be downloaded here:
www.cdc.gov/pneumococcal/hcp/vaccine-recommendations/app.html.
- **Special situations:**
 - **Age 19-49 years with certain underlying medical conditions or other risk factors who have** (NOTE: Underlying medical conditions or other risk factors include alcoholism, chronic heart/liver/lung disease, chronic kidney failure, cigarette smoking, cochlear implant, congenital or acquired asplenia, CSF leak, diabetes mellitus, generalized malignancy, HIV infection, Hodgkin disease, immunodeficiencies, iatrogenic immunosuppression, leukemia, lymphoma, multiple myeloma, nephrotic syndrome, solid organ transplant, or sickle cell disease or other hemoglobinopathies):
 - **Not previously received a PCV13, PCV15, PCV20, or PCV21 or whose previous vaccination history is unknown:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21.
 - If PCV15 is used, administer 1 dose PPSV23 at least 1 year after the PCV15 dose (may use minimum interval of 8 weeks for adults with an immunocompromising condition [NOTE: Immunocompromising conditions include chronic kidney failure, nephrotic syndrome, immunodeficiencies, iatrogenic immunosuppression, generalized malignancy, HIV infection, Hodgkin disease, leukemia, lymphoma, multiple myeloma, solid organ transplant, congenital or acquired asplenia, or sickle cell disease or other hemoglobinopathies], cochlear implant, or CSF leak).
 - **Previously received only PCV7:** Follow the recommendation above.
 - **Previously received only PCV13:** 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PCV13.
 - **Previously received only PPSV23:** 1 dose PCV15 or 1 dose PCV20 or PCV21, at least 1 year after the last PPSV23 dose.
 - If PCV15 is used, no additional PPSV23 doses are recommended.
 - **Previously received PCV13 and 1 dose of PPSV23:** 1 dose PCV20 or 1 dose PCV21 at least 5 years after the last pneumococcal vaccine dose.

- **Adults aged 19 years and older who have received PCV20 or PCV21:** No additional pneumococcal vaccine dose recommended.
- **Pregnancy:** No recommendation for PCV or PPSV23 due to limited data. Summary of existing data on pneumococcal vaccination during pregnancy can be found at www.cdc.gov/mmwr/volumes/72/rr/rr7203a1.htm.
- **PPSV23 not available:** Adults aged 19 years or older who received PCV15, but have not yet completed PPSV23 series, can complete the series with either 1 dose of PCV20 or 1 dose of PCV21 if they no longer have access to PPSV23.
- For guidance on determining which pneumococcal vaccines a patient needs and when, please refer to the mobile app which can be downloaded here: www.cdc.gov/pneumococcal/hcp/vaccine-recommendations/app.html.
- **Contraindications and precautions:**
 - For contraindications and precautions to Pneumococcal conjugate (PCV15 and PCV20), refer to [PCV Appendix](#); and for Pneumococcal polysaccharide (PPSV23), refer to [PPSV23 Appendix](#).

¶¶ Hepatitis A vaccination

- **Routine vaccination:**
 - **Any person who is not fully vaccinated and requests vaccination** (identification of risk factor not required): Complete 2-dose series HepA (Havrix 6-12 months apart or Vaqta 6-18 months apart [minimum interval: 6 months]) or 3-dose series HepA-HepB (Twinrix at 0, 1, 6 months [minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 5 months]).
- **Special situations:**
 - **Any person who is not fully vaccinated and who is at risk for hepatitis A virus infection or severe disease from hepatitis A virus infection:** Complete 2-dose series HepA or 3-dose series HepA-HepB as above. Risk factors include:
 - **Chronic liver disease** including persons with hepatitis B, hepatitis C, cirrhosis, fatty liver disease, alcoholic liver disease, autoimmune hepatitis, alanine aminotransferase (ALT) or aspartate aminotransferase (AST) level greater than twice the upper limit of normal.
 - **HIV infection.**
 - **Men who have sex with men (MSM).**
 - **Injection or noninjection drug use.**
 - **Persons experiencing homelessness.**
 - **Work with hepatitis A virus** in research laboratory or with nonhuman primates with hepatitis A virus infection.
 - **Travel in countries with high or intermediate endemic hepatitis A:** HepA-HepB (Twinrix) may be administered on an accelerated schedule of 3 doses at 0, 7, and 21-30 days, followed by a booster dose at 12 months.
 - **Close, personal contact with international adoptee** (eg, household or regular babysitting) in first 60 days after arrival from country with high or intermediate endemic hepatitis A: Dose 1 as soon as adoption is planned; preferably at least 2 weeks before adoptee's arrival.
 - **Pregnancy** if at risk for infection or severe outcome from infection during pregnancy.
 - **Settings for exposure,** including health care settings serving persons who use injection or noninjection drugs, or group homes and nonresidential day care facilities for developmentally disabled persons (individual risk factor screening not required).
- **Contraindications and precautions:**
 - For contraindications and precautions to hepatitis A (HepA) vaccination, refer to [HepA Appendix](#).

△△ Hepatitis B vaccination

- **Routine vaccination:**

- **Age 19 through 59 years:** Complete a 2- or 3- or 4-dose series.
 - 2-dose series only applies when 2 doses of Heplisav-B are used at least 4 weeks apart.
 - 3-dose series Engerix-B, PreHevbrio (NOTE: PreHevbrio is not recommended in pregnancy due to lack of safety data in pregnant persons), or Recombivax HB at 0, 1, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 8 weeks; dose 1 to dose 3: 16 weeks).
 - 3-dose series HepA-HepB (Twinrix) at 0, 1, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 5 months).
 - 4-dose series HepA-HepB (Twinrix) accelerated schedule of 3 doses at 0, 7, and 21-30 days, followed by a booster dose at 12 months.
- **Age 60 years or older without** known risk factors for hepatitis B virus infection **may** receive a HepB vaccine series.
- **Age 60 years or older with** known risk factors for hepatitis B virus infection **should** receive a HepB vaccine series.
- **Any adult age 60 years of age or older** who requests HepB vaccination **should** receive a HepB vaccine series.
 - **Risk factors for hepatitis B virus infection include:**
 - **Chronic liver disease** including persons with hepatitis C, cirrhosis, fatty liver disease, alcoholic liver disease, autoimmune hepatitis, ALT or AST level greater than twice the upper limit of normal.
 - **HIV infection.**
 - **Sexual exposure risk** (eg, sex partners of hepatitis B surface antigen [HBsAg]-positive persons, sexually active persons not in mutually monogamous relationships, persons seeking evaluation or treatment for a sexually transmitted infection, MSM).
 - **Current or recent injection drug use.**
 - **Percutaneous or mucosal risk for exposure to blood** (eg, household contacts of HBsAg-positive persons, residents and staff of facilities for developmentally disabled persons, health care and public safety personnel with reasonably anticipated risk for exposure to blood or blood-contaminated body fluids, persons on maintenance dialysis [including in-center or home hemodialysis and peritoneal dialysis], persons who are predialysis, and patients with diabetes [NOTE: **Age 60 years or older with diabetes:** Based on shared clinical decision making, 2-, 3-, or 4-dose series as above]).
 - **Incarceration.**
 - **Travel in countries with high or intermediate endemic hepatitis B.**
- **Special situations:**
 - **Patients on dialysis:** Complete a 3- or 4-dose series.
 - 3-dose series Recombivax HB at 0, 1, 6 months (NOTE: Use Dialysis Formulation 1 mL = 40 mcg).
 - 4-dose series Engerix-B at 0, 1, 2, and 6 months (NOTE: Use 2 mL dose instead of the normal adult dose of 1 mL).
 - **Age 20 years or older with an immunocompromising condition:** Complete a 2- or 3- or 4-dose series.
 - 3-dose series Recombivax HB at 0, 1, 6 months (NOTE: Use Dialysis Formulation 1 mL = 40 mcg).
 - 4-dose series Engerix-B at 0, 1, 2, and 6 months (NOTE: Use 2 mL dose instead of the normal adult dose of 1 mL).
 - 2-dose series Heplisav-B at 0, 1 months.
 - 3-dose series PreHevbrio at 0, 1, 6 months (NOTE: PreHevbrio is not recommended in pregnancy due to lack of safety data in pregnant persons).
- **Contraindications and precautions:**

- For contraindications and precautions to hepatitis B (HepB) vaccination, refer to [HepB Appendix](#).

◇◇ Meningococcal vaccination

■ Special situations for MenACWY:

- **Anatomical or functional asplenia (including sickle cell disease), HIV infection, persistent complement component deficiency, complement inhibitor (eg, eculizumab, ravulizumab) use:** 2-dose primary series Menveo or MenQuadfi at least 8 weeks apart; 1 booster dose 5 years after primary series and every 5 years if risk remains.
- **Travel in countries with hyperendemic or epidemic meningococcal disease, or for microbiologists routinely exposed to *Neisseria meningitidis*:** 1 dose Menveo or MenQuadfi; 1 booster dose 5 years after primary series and every 5 years if risk remains.
- **First-year college students who live in residential housing (if not previously vaccinated at age 16 years or older) or military recruits:** 1 dose Menveo or MenQuadfi.
- MenACWY vaccines may be administered simultaneously with MenB vaccines if indicated, but at a different anatomic site, if feasible.
- For MenACWY recommendations **in outbreak setting** (eg, in community or organizational settings, or among MSM) and **additional meningococcal vaccination** information, refer to www.cdc.gov/mmwr/volumes/69/rr/rr6909a1.htm.

■ Shared clinical decision-making for MenB:

- **Adolescents and young adults age 16-23 years (age 16-18 years preferred; NOTE: To optimize rapid protection [eg, for students starting college in less than 6 months], a 3-dose series [0, 1-2, 6 months] may be administered) not at increased risk for meningococcal disease:** Based on shared clinical decision-making.
 - **Bexsero or Trumenba (use same brand for all doses):** 2-dose series at least 6 months apart (if dose 2 is administered earlier than 6 months, administer dose 3 at least 4 months after dose 2)
- MenB vaccines may be administered simultaneously with MenACWY vaccines if indicated, but at a different anatomic site, if feasible.

■ Special situations for MenB:

- **Anatomical or functional asplenia (including sickle cell disease), persistent complement component deficiency, complement inhibitor (eg, eculizumab, ravulizumab) use, or microbiologists routinely exposed to *Neisseria meningitidis*.**
 - **Bexsero or Trumenba (use same brand for all doses including booster doses):** 3-dose primary series at 0, 1-2, 6 months (if dose 2 was administered at least 6 months after dose 1, dose 3 not needed; if dose 3 is administered earlier than 4 months after dose 2, a 4th dose should be administered at least 4 months after dose 3).
 - **Booster doses:** 1 booster dose 1 year after primary series and every 2-3 years if risk remains.
- **Pregnancy:** Delay MenB until after pregnancy due to lack of safety data in pregnant persons. May administer if at increased risk and vaccination benefits outweigh potential risks.
- For MenB recommendations **in outbreak setting** (eg, in community or organizational settings and among MSM) and **additional meningococcal vaccination information** refer to www.cdc.gov/mmwr/volumes/69/rr/rr6909a1.htm.
- MenB vaccines may be administered simultaneously with MenACWY vaccines if indicated, but at a different anatomic site, if feasible.
- Adults may receive a single dose of Penbraya (MenACWY-TT/MenB-FHbp) as an alternative to separate administration of MenACWY and MenB when both vaccines would be given on the same clinic day. For adults not at increased risk, if Penbraya is used for dose 1 MenB, then MenB-FHbp (Trumenba) should be administered for dose 2 MenB. For adults at increased risk of meningococcal disease, Penbraya may be used for additional MenACWY and MenB

doses (including booster doses) if both would be given on the same clinic day **and** at least 6 months have elapsed since most recent Penbraya dose.

▪ **Contraindications and precautions:**

- For contraindications and precautions to meningococcal ACWY (MenACWY) [MenACWY-CRM (Menveo); MenACWY-D (Menactra); MenACWY-TT (MenQuadfi)], refer to [MenACWY Appendix](#).
- For contraindications and precautions to meningococcal B (MenB) [MenB-4C (Bexsero); MenB-FHbp (Trumenba)], refer to [MenB Appendix](#).

§§ ***Haemophilus influenzae* type b vaccination**

▪ **Special situations:**

- **Anatomical or functional asplenia (including sickle cell disease):** 1 dose if previously did not receive Hib vaccine.
 - **Elective splenectomy:** 1 dose preferably at least 14 days before splenectomy.
- **Hematopoietic stem cell transplant (HSCT):** 3-dose series 4 weeks apart starting 6-12 months after successful transplant, regardless of Hib vaccination history.

▪ **Contraindications and precautions:**

- For contraindications and precautions to *Haemophilus influenzae* type b (Hib) vaccination, refer to [Hib Appendix](#).

¥¥ **Mpox vaccination**

▪ **Special situations:**

- **Any person at risk for mpox infection:** Complete 2-dose series, 28 days apart.

Risk factors for mpox infection include:

- Persons who are gay or bisexual and other MSM, transgender, or nonbinary people who in the past 6 months have had:
 - A new diagnosis of at least 1 sexually transmitted disease.
 - More than 1 sex partner.
 - Sex at a commercial sex venue.
 - Sex in association with a large public event in a geographic area where mpox transmission is occurring.
- Persons who are sexual partners of the persons described above.
- Persons who anticipate experiencing any of the situations described above.
- **Pregnancy:** There is currently no ACIP recommendation for Jynneos use in pregnancy due to lack of safety data in pregnant persons. Pregnant persons with any risk factor described above may receive Jynneos.
- **Health care personnel:** Vaccination to protect against occupational risk in health care settings is not routinely recommended.

▪ **Contraindications and precautions:**

- For contraindications and precautions to mpox, refer to [Mpox Appendix](#).

‡‡ **Poliovirus vaccination**

▪ **Routine vaccination:**

- **Adults known or suspected to be unvaccinated or incompletely vaccinated:** Administer remaining doses (1, 2, or 3 inactivated poliovirus [IPV] doses) to complete a 3-dose primary series (NOTE: Complete primary series consists of at least 3 doses of IPV or trivalent oral poliovirus vaccine [tOPV] in any combination). Unless there are specific reasons to believe they were not vaccinated, most adults who were born and raised in the United States can assume they were vaccinated against polio as children.

▪ **Special situations:**

- **Adults at increased risk of exposure to poliovirus who completed primary series** (NOTE: Complete primary series consists of at least 3 doses of IPV or tOPV in any combination): May administer one lifetime IPV booster.

- For detailed information, refer to www.cdc.gov/vaccines/vpd/polio/hcp/recommendations.html.
- **Contraindications and precautions:**
 - For contraindications and precautions to Poliovirus vaccine, IPV, refer to [Poliovirus Appendix](#).

Reproduced from: Advisory Committee on Immunization Practices. Adult Immunization Schedule by Age, Recommendations for Ages 19 Years or Older, United States, 2025. Centers for Disease Control and Prevention. Available at: <https://www.cdc.gov/vaccines/hcp/imz-schedules/adult-age.html> (Accessed on December 10, 2024).

Figure 2: Adult immunization schedule by medical condition and other indication – Recommendations for ages 19 years or older, United States, 2025

	Indication										
Vaccine	Pregnancy	Immuno-compromised (excluding HIV infection)	HIV infection CD4 percentage and count		Men who have sex with men	Asplenia, complement deficiency	Heart or lung disease	Kidney failure, End-stage renal disease; or on dialysis	Chronic liver disease; alcoholism *	Diabetes	Health care personnel Δ,‡,†,◇◇
			<15% or <200 mm ³	≥15% and ≥200 mm ³							
COVID-19¶		Refer to footnotes									
Influenza inactivated Influenza recombinantΔ		Solid organ transplant (refer to footnotes)	1 dose annually								
Influenza live, attenuated (LAIV4)Δ					1 dose annually if age 19-49 years		1 dose annually if age 19-49 years				
Respiratory syncytial virus (RSV)◇	Seasonal administration (refer to footnotes)	Refer to footnotes				Refer to footnotes			Liver disease (refer to footnotes)	Refer to footnotes	
Tetanus, diphtheria, pertussis (Tdap or Td)§	Tdap: 1 dose each pregnancy	1 dose Tdap, then Td or Tdap booster every 10 years									
Measles, mumps, rubella (MMR)‡	***										
Varicella (VAR)†	***	Refer to footnotes									
Zoster recombinant (RZV)†		Refer to footnotes									
Human papillomavirus (HPV)**	***	3-dose series if indicated									
Pneumococcal (PCV15, PCV20, PCV21, PPSV23)¶¶											
Hepatitis A (HepA)ΔΔ											
Hepatitis B (HepB)◇◇	Refer to footnotes								Age ≥60 years		
Meningococcal A, C, W, Y (MenACWY)§§											
Meningococcal B (MenB)§§											
Haemophilus influenzae type b (Hib)‡‡		HSCT: 3 doses				Asplenia: 1 dose					
Mpox**	Refer to footnotes				Refer to footnotes						Refer to footnotes
Inactivated poliovirus (IPV)††		Complete 3-dose series if incompletely vaccinated. Self-report of previous doses acceptable. (Refer to footnotes.)									

 Recommended for all adults who lack documentation of vaccination, or lack evidence of immunity	 Not recommended for all adults, but recommended for some adults based on either age or increased risk or for severe outcomes from disease	 Recommended based on shared clinical decision-making
 Recommended for all adults, and additional doses may be necessary based on medical condition or other indications. Refer to footnotes.	 Precaution: Might be indicated if benefit of protection outweighs risk of adverse reaction	 Contraindicated or not recommended
 No guidance/not applicable		

Administer recommended vaccines if vaccination history is incomplete or unknown. Do not restart or add doses to vaccine series if there are extended intervals between doses. The use of trade names is for identification purposes only and does not imply endorsement by the ACIP or CDC.

NOTES

For vaccine recommendations for persons 18 years of age or younger, refer to the [Recommended Child and Adolescent Immunization Schedule](#).

Additional information

- For calculating intervals between doses, 4 weeks = 28 days. Intervals of ≥ 4 months are determined by calendar months.
- Within a number range (eg, 12-18), a dash (-) should be read as "through."
- Vaccine doses administered ≤ 4 days before the minimum age or interval are considered valid. Doses of any vaccine administered ≥ 5 days earlier than the minimum age or minimum interval should not be counted as valid and should be repeated. **The repeat dose should be spaced after the invalid dose by the recommended minimum interval.** For further details, refer to Table 3-2, Recommended and minimum ages and intervals between vaccine doses, in [Timing and Spacing of Immunobiologics](#).
- Information on travel vaccination requirements and recommendations is available at [cdc.gov/travel/](https://www.cdc.gov/travel/).
- For vaccination of persons with immunodeficiencies, refer to Table 8-1, Vaccination of persons with primary and secondary immunodeficiencies, in [Altered Immunocompetence](#).
- For information about vaccination in the setting of a vaccine-preventable disease outbreak, contact your state or local health department.
- The National Vaccine Injury Compensation Program (VICP) is a no-fault alternative to the traditional legal system for resolving vaccine injury claims. All vaccines included in the adult immunization schedule except PPSV23, RSV, RZV, mpox, and COVID-19 vaccines are covered by the National Vaccine Injury Compensation Program (VICP). Mpox and COVID-19 vaccines are covered by the Countermeasures Injury Compensation Program (CICP). For more information, refer to www.hrsa.gov/vaccinecompensation or www.hrsa.gov/cicp.

HSCT: hematopoietic stem cell transplant.

* Precaution for LAIV4 does not apply to alcoholism.

¶ COVID-19 vaccination

- **Routine vaccination:**
 - **Age 19-64 years.**
 - **Unvaccinated:**
 - 1 dose 2024-25 Moderna or Pfizer-BioNTech.
 - 2 doses 2024-25 Novavax at 0, 3-8 weeks.
 - **Previously vaccinated before 2024-25 vaccine with:**
 - **1 or more doses Moderna or Pfizer-BioNTech:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech at least 8 weeks after the most recent dose.
 - **1 dose Novavax:** 1 dose 2024-25 Novavax 3-8 weeks after most recent dose. If more than 8 weeks after the most recent dose, administer 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech.
 - **2 or more doses Novavax:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech at least 8 weeks after the most recent dose.

- **1 or more doses Janssen:** 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech.
- **Age 65 years and older.**
 - **Unvaccinated:** Follow recommendations above for unvaccinated persons ages 19-64 years **and** administer dose 2 of 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months).
 - **Previously vaccinated before 2024-25 vaccine:** Follow recommendations above for previously vaccinated persons ages 19-64 years **and** administer dose 2 of 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months).
- **Special situations:**
 - **Persons who are moderately or severely immunocompromised. Use vaccine from the same manufacturer for all doses in the initial vaccination series.**
 - **Unvaccinated:**
 - 4 doses (3-dose initial series 2024-25 Moderna at 0, 4 weeks, and at least 4 weeks after dose 2, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses. ¶¶¶
 - 4 doses (3-dose initial series 2024-25 Pfizer-BioNTech at 0, 3 weeks, and at least 4 weeks after dose 2, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses. ¶¶¶
 - 3 doses (2-dose initial series 2024-25 Novavax at 0, 3 weeks, followed by 1 dose Moderna or Novavax or Pfizer-BioNTech 6 months later [minimum interval 2 months]). May administer additional doses. ¶¶¶
 - **Incomplete initial vaccination series before 2024-25 vaccine:**
 - **Previous vaccination with Moderna**
 - **1 dose Moderna:** Complete initial series with 2 doses 2024-25 Moderna at least 4 weeks apart (administer dose 1 4 weeks after most recent dose), followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses. ¶¶¶
 - **2 doses Moderna:** Complete initial series with 1 dose 2024-25 Moderna at least 4 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses. ¶¶¶
 - **Previous vaccination with Pfizer-BioNTech**
 - **1 dose Pfizer-BioNTech:** Complete initial series with 2 doses 2024-25 Pfizer-BioNTech at least 4 weeks apart (administer dose 1 3 weeks after most recent dose), followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses. ¶¶¶
 - **2 doses Pfizer-BioNTech:** Complete initial series with 1 dose 2024-25 Pfizer-BioNTech at least 4 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses. ¶¶¶
 - **Previous vaccination with Novavax**
 - **1 dose Novavax:** Complete initial series with 1 dose 2024-25 Novavax at least 3 weeks after most recent dose, followed by 1 dose 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months later (minimum interval 2 months). May administer additional doses. ¶¶¶
 - **Completed the initial vaccination series before 2024-25 vaccine with:**
 - **3 or more doses Moderna or 3 or more doses Pfizer-BioNTech:** 2 doses 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months apart (minimum interval 2 months).

Administer dose 1 at least 8 weeks after the most recent dose. May administer additional doses. ¶¶¶

- **2 or more doses Novavax:** 2 doses 2024-25 Moderna or Novavax or Pfizer-BioNTech 6 months apart (minimum interval 2 months). Administer dose 1 at least 8 weeks after the most recent dose. May administer additional doses. ¶¶¶

- ¶¶¶ **Additional doses of 2024-25 COVID-19 vaccine for moderately or severely immunocompromised:**

- Based on shared clinical decision-making and administered at least 2 months after the most recent dose (refer to Table 2 at www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#table-02). For description of moderate and severe immunocompromising conditions and treatment, refer to www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#immunocompromising-conditions-treatment.
- Unvaccinated persons have never received any COVID-19 vaccine doses. There is no preferential recommendation for the use of one COVID-19 vaccine over another when more than one recommended age-appropriate vaccine is available. Administer an age-appropriate COVID-19 vaccine product for each dose.
- For more information about the interchangeability of COVID-19 vaccines, refer to <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/interim-considerations-us.html#Interchangeability>.
- Current COVID-19 schedule and dose formulation available at www.cdc.gov/covidschedule.

- **Contraindications and precautions:**

- For contraindications and precautions to COVID-19 vaccination, refer to [COVID-19 Appendix](#).

Δ Influenza vaccination

- **Routine vaccination:**

- **Age 19 years or older:** 1 dose any influenza vaccine appropriate for age and health status annually.
 - **Solid organ transplant recipients aged 19 through 64 years receiving immunosuppressive medications:** HD-IIV3 and aIIV3 are acceptable options. No preference over other age-appropriate IIV3 or RIV3.
 - **Age 65 years or older:** Any one of HD-IIV3, RIV3, or aIIV3 is preferred. If none of these three vaccines are available, then any other age-appropriate influenza vaccine should be used.
- For the 2024-25 season, refer to www.cdc.gov/mmwr/volumes/73/rr/rr7305a1.htm.
- For the 2025-26 season, refer to the 2025-26 ACIP influenza vaccine recommendations.

- **Special situations:**

- **Close contacts (eg, caregivers, health care workers) of severely immunosuppressed persons who require a protected environment:** Should not receive LAIV3. If LAIV3 is given, they should avoid contact with/caring for such immunosuppressed persons for 7 days after vaccination.
- **NOTE:** Persons with an egg allergy can receive any influenza vaccine (egg-based and non-egg-based) appropriate for age and health status.

- **Contraindications and precautions:**

- For contraindications and precautions to influenza vaccination, refer to [IIV3 Appendix](#), [LAIV3 Appendix](#), [ccIIV3 Appendix](#), and [RIV3 Appendix](#).

◇ Respiratory syncytial virus vaccination

- **Routine vaccination:**

- **Pregnant persons of any age.**

- **Pregnant at 32 weeks 0 days through 36 weeks and 6 days gestation from September through January in most of the continental United States** (NOTE: Providers in jurisdictions with RSV seasonality that differs from most of the continental United States [eg, Alaska, jurisdiction with tropical climate] should follow guidance from public health authorities on timing of administration. Refer to the 2025 Child and Adolescent Immunization Schedule for considerations regarding nirsevimab administration to infants): 1 dose Abrysvo. Administer RSV vaccine regardless of previous RSV infection.
- Either maternal RSV vaccination with Abrysvo or infant immunization with nirsevimab (RSV monoclonal antibody) is recommended to prevent severe respiratory syncytial virus disease in infants.
- **All other pregnant persons:** RSV vaccine not recommended.
- **Subsequent pregnancies:** Additional doses not recommended. No data are available to inform whether additional doses are needed in subsequent pregnancies. Infants born to pregnant persons who received RSV vaccine during a previous pregnancy should receive nirsevimab.
- **Age 75 years or older.**
 - **Unvaccinated:** 1 dose (Arexvy or Abrysvo or mResvia). Additional doses not recommended.
 - **Previously vaccinated:** Additional doses not recommended. No data are available to inform whether additional doses are needed.
- **Special situations:**
 - **Age 60-74 years.**
 - **Unvaccinated and at increased risk of severe RSV disease^{ΔΔΔ}:** 1 dose (Arexvy or Abrysvo or mResvia). Additional doses not recommended.
 - **Previously vaccinated:** Additional doses not recommended. No data are available to inform whether additional doses are needed.
 - Persons 60 years and older can get RSV vaccine at any time but best to administer in late summer and early fall before RSV spreads in communities—ideally August through October in most of continental United States. For further guidance, refer to www.cdc.gov/mmwr/volumes/73/wr/mm7332e1.htm.
 - **ΔΔΔ People can self-attest to the presence of a risk factor. The following medical and other conditions increase the risk of severe RSV disease:**
 - Chronic cardiovascular disease (eg, heart failure, coronary artery disease, congenital heart disease). Excludes isolated hypertension.
 - Chronic lung or respiratory disease (eg, chronic obstructive pulmonary disease, emphysema, asthma, interstitial lung disease, cystic fibrosis).
 - End-stage renal disease, dependence on hemodialysis, or other renal replacement therapy.
 - Diabetes mellitus complicated by chronic kidney disease, neuropathy, retinopathy, or other end-organ damage.
 - Diabetes mellitus requiring treatment with insulin or sodium-glucose cotransporter 2 (SGLT2) inhibitor.
 - Neurologic or neuromuscular conditions causing impaired airway clearance or respiratory muscle weakness (eg, post-stroke dysphagia, amyotrophic lateral sclerosis, muscular dystrophy). Excludes history of stroke without impaired airway clearance.
 - Chronic liver disease (eg, cirrhosis).
 - Chronic hematologic conditions (eg, sickle cell disease, thalassemia).
 - Severe obesity (body mass index ≥ 40 kg/m²).
 - Moderate or severe immune compromise.
 - Residence in a nursing home.

- Other chronic medical conditions or risk factors that a health care provider determines would increase the risk of severe disease due to viral respiratory infection (eg, frailty, concern for presence of undiagnosed chronic medical conditions, residence in a remote or rural community where escalation of medical care is challenging).
 - **Contraindications and precautions:**
 - For contraindications and precautions to RSV vaccine, refer to [RSV Appendix](#).
-

§ Tetanus, diphtheria, and pertussis vaccination

- **Routine vaccination:**
 - **Completed primary series and received at least 1 dose Tdap at age 10 years or older:** Td or Tdap every 10 years thereafter.
 - **Completed primary series and did NOT receive Tdap at age 10 years or older:** 1 dose Tdap, then Td or Tdap every 10 years thereafter.
 - **Unvaccinated or incomplete primary vaccination series for tetanus, diphtheria, or pertussis:** Administer remaining doses (1, 2, or 3 doses) to complete 3-dose primary series. 1 dose Tdap followed by 1 dose Td or Tdap at least 4 weeks later, and a third dose of Td or Tdap 6-12 months later (Tdap is preferred as first dose and can be substituted for any Td dose), then Td or Tdap every 10 years thereafter.
- **Special situations:**
 - **Pregnancy:** 1 dose Tdap during each pregnancy, preferably in early part of gestational weeks 27-36.
 - **Wound management:** Persons with 3 or more doses of tetanus-toxoid-containing vaccine: For clean and minor wounds, administer Tdap or Td if more than 10 years since last dose of tetanus-toxoid-containing vaccine; for all other wounds, administer Tdap or Td if more than 5 years since last dose of tetanus-toxoid-containing vaccine. Tdap is preferred for persons who have not previously received Tdap or whose Tdap history is unknown. If a tetanus-toxoid-containing vaccine is indicated for a pregnant person, use Tdap. For detailed information, refer to www.cdc.gov/mmwr/volumes/69/wr/mm6903a5.htm.
- **Contraindications and precautions:**
 - For contraindications and precautions to tetanus, diphtheria, and acellular pertussis (Tdap), refer to [Tdap Appendix](#).

¥ Measles, mumps, and rubella vaccination

- **Routine vaccination:**
 - **No evidence of immunity to measles, mumps, or rubella:** 1 dose.
 - **Evidence of immunity:** Born before 1957 (except for health care personnel, refer below), documentation of receipt of MMR vaccine, laboratory evidence of immunity or disease (diagnosis of disease without laboratory confirmation is not evidence of immunity).
- **Special situations:**
 - **Pregnancy with no evidence of immunity to rubella:** MMR contraindicated during pregnancy; after pregnancy (before discharge from health care facility), 1 dose.
 - **Nonpregnant persons of childbearing age with no evidence of immunity to rubella:** 1 dose.
 - **Human immunodeficiency virus (HIV) infection with CD4 percentages $\geq 15\%$ and CD4 count ≥ 200 cells/mm³ for at least 6 months and no evidence of immunity to measles, mumps, or rubella:** Complete 2-dose series at least 4 weeks apart; MMR contraindicated for HIV infection with CD4 percentage $< 15\%$ or CD4 count < 200 cells/mm³.
 - **Severe immunocompromising conditions:** MMR contraindicated.
 - **Students in postsecondary educational institutions, international travelers, and household or close, personal contacts of immunocompromised persons with no**

evidence of immunity to measles, mumps, or rubella: Complete 2-dose series at least 4 weeks apart if previously did not receive any doses of MMR or 1 dose if previously received 1 dose MMR.

- **In mumps outbreak settings,** for information about additional doses of MMR (including 3rd dose of MMR), refer to www.cdc.gov/mmwr/volumes/67/wr/mm6701a7.htm.
- **Health care personnel:**
 - **Born before 1957 with no evidence of immunity to measles, mumps, or rubella:** Consider 2-dose series at least 4 weeks apart for protection against measles or mumps or 1 dose for protection against rubella.
 - **Born in 1957 or later with no evidence of immunity to measles, mumps, or rubella:** Complete 2-dose series at least 4 weeks apart for protection against measles or mumps or at least 1 dose for protection against rubella.
- **Contraindications and precautions:**
 - For contraindications and precautions to measles, mumps, rubella (MMR) vaccine, refer to [MMR Appendix](#).

‡ **Varicella vaccination**

- **Routine vaccination:**
 - **No evidence of immunity to varicella:** 2-dose series 4-8 weeks apart if previously did not receive varicella-containing vaccine (VAR or measles-mumps-rubella-varicella vaccine [MMRV] for children); if previously received 1 dose varicella-containing vaccine, 1 dose at least 4 weeks after first dose.
 - **Evidence of immunity:** United States-born before 1980 (except for pregnant persons and health care personnel [refer below]), documentation of 2 doses varicella-containing vaccine at least 4 weeks apart, diagnosis or verification of history of varicella or herpes zoster by a health care provider, laboratory evidence of immunity or disease.
- **Special situations:**
 - **Pregnancy with no evidence of immunity to varicella:** VAR contraindicated during pregnancy; after pregnancy (before discharge from health care facility), 1 dose if previously received 1 dose varicella-containing vaccine or dose 1 of 2-dose series (dose 2: 4 to 8 weeks later) if previously did not receive any varicella-containing vaccine, regardless of whether United States-born before 1980.
 - **Health care personnel with no evidence of immunity to varicella:** 1 dose if previously received 1 dose varicella-containing vaccine; 2-dose series 4-8 weeks apart if previously did not receive any varicella-containing vaccine, regardless of whether United States-born before 1980.
 - **HIV infection with CD4 percentages $\geq 15\%$ and CD4 count ≥ 200 cells/mm³ with no evidence of immunity:** Vaccination may be considered (2 doses 3 months apart); VAR contraindicated for HIV infection with CD4 percentage $< 15\%$ or CD4 count < 200 cells/mm³.
 - **Severe immunocompromising conditions:** VAR contraindicated.
- **Contraindications and precautions:**
 - For contraindications and precautions to varicella (VAR) vaccine, refer to [VAR Appendix](#).

† **Zoster vaccination**

- **Routine vaccination:**
 - **Age 50 years or older** (NOTE: Serologic evidence of prior varicella is not necessary for zoster vaccination. However, if serologic evidence of varicella susceptibility becomes available, providers should follow ACIP guidelines for varicella vaccination first. RZV is not indicated for the prevention of varicella, and there are limited data on the use of RZV in persons without a history of varicella or varicella vaccination): 2-dose series recombinant zoster vaccine (RZV; Shingrix) 2-6 months apart (minimum interval: 4 weeks; repeat dose if administered too

soon), regardless of previous herpes zoster or history of zoster vaccine live (ZVL; Zostavax) vaccination.

▪ **Special situations:**

- **Pregnancy:** There is currently no ACIP recommendation for RZV use in pregnancy. Consider delaying RZV until after pregnancy.
- **Immunocompromising conditions (including persons with HIV regardless of CD4 count;** NOTE: If there is no documented history of varicella, varicella vaccination, or herpes zoster, providers should refer to the clinical considerations for use of RZV in immunocompromised adults aged ≥19 years and the ACIP varicella vaccine recommendations for further guidance: www.cdc.gov/mmwr/volumes/71/wr/mm7103a2.htm): 2-dose series recombinant zoster vaccine (RZV; Shingrix) 2-6 months apart (minimum interval: 4 weeks; repeat dose if administered too soon). For detailed information, refer to www.cdc.gov/shingles/vaccination/immunocompromised-adults.html.

▪ **Contraindications and precautions:**

- For contraindications and precautions to zoster recombinant vaccine (RZV), refer to [RZV Appendix](#).

**** Human papillomavirus vaccination**

▪ **Routine vaccination:**

- **All persons through age 26 years:** Complete 2- or 3-dose series depending on age at initial vaccination or condition.
 - **Age 9-14 years at initial vaccination and received 1 dose or 2 doses less than 5 months apart:** 1 additional dose.
 - **Age 9-14 years at initial vaccination and received 2 doses at least 5 months apart:** HPV vaccination series complete, no additional dose needed.
 - **Age 15 years or older at initial vaccination:** 3-dose series at 0, 1-2 months, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 12 weeks; dose 1 to dose 3: 5 months; repeat dose if administered too soon).
- No additional dose recommended when any HPV vaccine series of any valency has been completed using the recommended dosing intervals.

▪ **Shared clinical decision-making:**

- **Adults age 27-45 years:** Based on shared clinical decision-making, complete a 2-dose series (if initiated age 9-14 years) or 3-dose series (if initiated ≥15 years).
- For additional information on shared clinical decision-making for HPV; refer to www.cdc.gov/vaccines/hcp/admin/downloads/isd-job-aid-scdm-hpv-shared-clinical-decision-making-hpv.pdf.

▪ **Special situations:**

- **Age ranges recommended above for routine and catch-up vaccination or shared clinical decision-making also apply in special situations.**
 - **Immunocompromising conditions, including HIV infection:** Complete 3-dose series, even for those who initiate vaccination at age 9 through 14 years.
 - **Pregnancy:** Pregnancy testing is not needed before vaccination. HPV vaccination is not recommended until after pregnancy. No intervention needed if inadvertently vaccinated while pregnant.

▪ **Contraindications and precautions:**

- For contraindications and precautions to human papillomavirus (HPV) vaccination, refer to [HPV Appendix](#).

¶¶ Pneumococcal vaccination

▪ **Routine vaccination:**

- **Age 50 years or older who have:**

- **Not previously received a dose of PCV13, PCV15, PCV20, or PCV21 or whose previous vaccination history is unknown:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21.
 - If PCV15 is used, administer 1 dose PPSV23 at least 1 year after the PCV15 dose (may use minimum interval of 8 weeks for adults with an immunocompromising condition [NOTE: Immunocompromising conditions include chronic renal failure, nephrotic syndrome, immunodeficiencies, iatrogenic immunosuppression, generalized malignancy, HIV infection, Hodgkin disease, leukemia, lymphoma, multiple myeloma, solid organ transplant, congenital or acquired asplenia, or sickle cell disease or other hemoglobinopathies], cochlear implant, or cerebrospinal fluid [CSF] leak).
- **Previously received only PCV7:** Follow the recommendation above.
- **Previously received only PCV13:** 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PCV13 dose.
- **Previously received only PPSV23:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PPSV23 dose.
 - If PCV15 is used, no additional PPSV23 doses are recommended.
- **Previously received both PCV13 and PPSV23 but NO PPSV23 was received at age 65 years or older:** 1 dose PCV20 or 1 dose PCV21 at least 5 years after the last pneumococcal vaccine dose.
- **Previously received both PCV13 and PPSV23, AND PPSV23 was received at age 65 years or older:** Based on shared clinical decision-making, 1 dose of PCV20 or 1 dose of PCV21 at least 5 years after the last pneumococcal vaccine dose.
- For guidance on determining which pneumococcal vaccines a patient needs and when, please refer to the mobile app which can be downloaded here:
www.cdc.gov/pneumococcal/hcp/vaccine-recommendations/app.html.
- **Special situations:**
 - **Age 19-49 years with certain underlying medical conditions or other risk factors who have** (NOTE: Underlying medical conditions or other risk factors include alcoholism, chronic heart/liver/lung disease, chronic renal failure, cigarette smoking, cochlear implant, congenital or acquired asplenia, CSF leak, diabetes mellitus, generalized malignancy, HIV infection, Hodgkin disease, immunodeficiencies, iatrogenic immunosuppression, leukemia, lymphoma, multiple myeloma, nephrotic syndrome, solid organ transplant, or sickle cell disease or other hemoglobinopathies):
 - **Not previously received a PCV13, PCV15, PCV20, or PCV21 or whose previous vaccination history is unknown:** 1 dose PCV15 or 1 dose PCV20 or 1 dose PCV21.
 - If PCV15 is used, administer 1 dose PPSV23 at least 1 year after the PCV15 dose (may use minimum interval of 8 weeks for adults with an immunocompromising condition [NOTE: Immunocompromising conditions include chronic renal failure, nephrotic syndrome, immunodeficiencies, iatrogenic immunosuppression, generalized malignancy, HIV infection, Hodgkin disease, leukemia, lymphoma, multiple myeloma, solid organ transplant, congenital or acquired asplenia, or sickle cell disease or other hemoglobinopathies], cochlear implant, or CSF leak).
 - **Previously received only PCV7:** Follow the recommendation above.
 - **Previously received only PCV13:** 1 dose PCV20 or 1 dose PCV21 at least 1 year after the last PCV13.
 - **Previously received only PPSV23:** 1 dose PCV15 or 1 dose PCV20 or PCV21, at least 1 year after the last PPSV23 dose.
 - If PCV15 is used, no additional PPSV23 doses are recommended.
 - **Previously received PCV13 and 1 dose of PPSV23:** 1 dose PCV20 or 1 dose PCV21 at least 5 years after the last pneumococcal vaccine dose.

- **Adults aged 19 years and older who have received PCV20 or PCV21:** No additional pneumococcal vaccine dose recommended.
- **Pregnancy:** No recommendation for PCV or PPSV23 due to limited data. Summary of existing data on pneumococcal vaccination during pregnancy can be found at www.cdc.gov/mmwr/volumes/72/rr/rr7203a1.htm.
- **PPSV23 not available:** Adults aged 19 years or older who received PCV15, but have not yet completed PPSV23 series, can complete the series with either 1 dose of PCV20 or 1 dose of PCV21 if they no longer have access to PPSV23.
- For guidance on determining which pneumococcal vaccines a patient needs and when, please refer to the mobile app which can be downloaded here: www.cdc.gov/pneumococcal/hcp/vaccine-recommendations/app.html.
- **Contraindications and precautions:**
 - For contraindications and precautions to Pneumococcal conjugate (PCV15 and PCV20), refer to [PCV Appendix](#); and for Pneumococcal polysaccharide (PPSV23), refer to [PPSV23 Appendix](#).

△△ Hepatitis A vaccination

- **Routine vaccination:**
 - **Any person who is not fully vaccinated and requests vaccination** (identification of risk factor not required): Complete 2-dose series HepA (Havrix 6-12 months apart or Vaqta 6-18 months apart [minimum interval: 6 months]) or 3-dose series HepA-HepB (Twinrix at 0, 1, 6 months [minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 5 months]).
- **Special situations:**
 - **Any person who is not fully vaccinated and who is at risk for hepatitis A virus infection or severe disease from hepatitis A virus infection:** Complete 2-dose series HepA or 3-dose series HepA-HepB as above. Risk factors include:
 - **Chronic liver disease** including persons with hepatitis B, hepatitis C, cirrhosis, fatty liver disease, alcoholic liver disease, autoimmune hepatitis, alanine aminotransferase (ALT) or aspartate aminotransferase (AST) level greater than twice the upper limit of normal.
 - **HIV infection.**
 - **Men who have sex with men (MSM).**
 - **Injection or noninjection drug use.**
 - **Persons experiencing homelessness.**
 - **Work with hepatitis A virus** in research laboratory or with nonhuman primates with hepatitis A virus infection.
 - **Travel in countries with high or intermediate endemic hepatitis A:** HepA-HepB (Twinrix) may be administered on an accelerated schedule of 3 doses at 0, 7, and 21-30 days, followed by a booster dose at 12 months.
 - **Close, personal contact with international adoptee** (eg, household or regular babysitting) in first 60 days after arrival from country with high or intermediate endemic hepatitis A: Dose 1 as soon as adoption is planned; preferably at least 2 weeks before adoptee's arrival.
 - **Pregnancy** if at risk for infection or severe outcome from infection during pregnancy.
 - **Settings for exposure,** including health care settings serving persons who use injection or noninjection drugs, or group homes and nonresidential day care facilities for developmentally disabled persons (individual risk factor screening not required).
- **Contraindications and precautions:**
 - For contraindications and precautions to hepatitis A (HepA) vaccination, refer to [HepA Appendix](#).

◇◇ Hepatitis B vaccination

- **Routine vaccination:**

- **Age 19 through 59 years:** Complete a 2- or 3- or 4-dose series.
 - 2-dose series only applies when 2 doses of Heplisav-B are used at least 4 weeks apart.
 - 3-dose series Engerix-B, PreHevbrio (NOTE: PreHevbrio is not recommended in pregnancy due to lack of safety data in pregnant persons), or Recombivax HB at 0, 1, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 8 weeks; dose 1 to dose 3: 16 weeks).
 - 3-dose series HepA-HepB (Twinrix) at 0, 1, 6 months (minimum intervals: dose 1 to dose 2: 4 weeks; dose 2 to dose 3: 5 months).
 - 4-dose series HepA-HepB (Twinrix) accelerated schedule of 3 doses at 0, 7, and 21-30 days, followed by a booster dose at 12 months.
- **Age 60 years or older without** known risk factors for hepatitis B virus infection **may** receive a HepB vaccine series.
- **Age 60 years or older with** known risk factors for hepatitis B virus infection **should** receive a HepB vaccine series.
- **Any adult age 60 years of age or older** who requests HepB vaccination **should** receive a HepB vaccine series.
 - **Risk factors for hepatitis B virus infection include:**
 - **Chronic liver disease** including persons with hepatitis C, cirrhosis, fatty liver disease, alcoholic liver disease, autoimmune hepatitis, ALT or AST level greater than twice the upper limit of normal.
 - **HIV infection.**
 - **Sexual exposure risk** (eg, sex partners of hepatitis B surface antigen [HBsAg]-positive persons, sexually active persons not in mutually monogamous relationships, persons seeking evaluation or treatment for a sexually transmitted infection, MSM).
 - **Current or recent injection drug use.**
 - **Percutaneous or mucosal risk for exposure to blood** (eg, household contacts of HBsAg-positive persons, residents and staff of facilities for developmentally disabled persons, health care and public safety personnel with reasonably anticipated risk for exposure to blood or blood-contaminated body fluids, persons on maintenance dialysis [including in-center or home hemodialysis and peritoneal dialysis], persons who are predialysis, and patients with diabetes [NOTE: **Age 60 years or older with diabetes:** Based on shared clinical decision making, 2-, 3-, or 4-dose series as above]).
 - **Incarceration.**
 - **Travel in countries with high or intermediate endemic hepatitis B.**
- **Special situations:**
 - **Patients on dialysis:** Complete a 3- or 4-dose series.
 - 3-dose series Recombivax HB at 0, 1, 6 months (NOTE: Use Dialysis Formulation 1 mL = 40 mcg).
 - 4-dose series Engerix-B at 0, 1, 2, and 6 months (NOTE: Use 2 mL dose instead of the normal adult dose of 1 mL).
 - **Age 20 years or older with an immunocompromising condition:** Complete a 2- or 3- or 4-dose series.
 - 3-dose series Recombivax HB at 0, 1, 6 months (NOTE: Use Dialysis Formulation 1 mL = 40 mcg).
 - 4-dose series Engerix-B at 0, 1, 2, and 6 months (NOTE: Use 2 mL dose instead of the normal adult dose of 1 mL).
 - 2-dose series Heplisav-B at 0, 1 months.
 - 3-dose series PreHevbrio at 0, 1, 6 months (NOTE: PreHevbrio is not recommended in pregnancy due to lack of safety data in pregnant persons).
- **Contraindications and precautions:**

- For contraindications and precautions to hepatitis B (HepB) vaccination, refer to [HepB Appendix](#).

§§ Meningococcal vaccination

■ Special situations for MenACWY:

- **Anatomical or functional asplenia (including sickle cell disease), HIV infection, persistent complement component deficiency, complement inhibitor (eg, eculizumab, ravulizumab) use:** 2-dose primary series Menveo or MenQuadfi at least 8 weeks apart; 1 booster dose 5 years after primary series and every 5 years if risk remains.
- **Travel in countries with hyperendemic or epidemic meningococcal disease, or for microbiologists routinely exposed to *Neisseria meningitidis*:** 1 dose Menveo or MenQuadfi; 1 booster dose 5 years after primary series and every 5 years if risk remains.
- **First-year college students who live in residential housing (if not previously vaccinated at age 16 years or older) or military recruits:** 1 dose Menveo or MenQuadfi.
- MenACWY vaccines may be administered simultaneously with MenB vaccines if indicated, but at a different anatomic site, if feasible.
- For MenACWY recommendations **in outbreak setting** (eg, in community or organizational settings, or among MSM) and **additional meningococcal vaccination** information, refer to www.cdc.gov/mmwr/volumes/69/rr/rr6909a1.htm.

■ Shared clinical decision-making for MenB:

- **Adolescents and young adults age 16-23 years (age 16-18 years preferred; NOTE: To optimize rapid protection [eg, for students starting college in less than 6 months], a 3-dose series [0, 1-2, 6 months] may be administered) not at increased risk for meningococcal disease:** Based on shared clinical decision-making.
 - **Bexsero or Trumenba (use same brand for all doses):** 2-dose series at least 6 months apart (if dose 2 is administered earlier than 6 months, administer dose 3 at least 4 months after dose 2)
- MenB vaccines may be administered simultaneously with MenACWY vaccines if indicated, but at a different anatomic site, if feasible.

■ Special situations for MenB:

- **Anatomical or functional asplenia (including sickle cell disease), persistent complement component deficiency, complement inhibitor (eg, eculizumab, ravulizumab) use, or microbiologists routinely exposed to *Neisseria meningitidis*.**
 - **Bexsero or Trumenba (use same brand for all doses including booster doses):** 3-dose primary series at 0, 1-2, 6 months (if dose 2 was administered at least 6 months after dose 1, dose 3 not needed; if dose 3 is administered earlier than 4 months after dose 2, a 4th dose should be administered at least 4 months after dose 3).
 - **Booster doses:** 1 booster dose 1 year after primary series and every 2-3 years if risk remains.
- **Pregnancy:** Delay MenB until after pregnancy due to lack of safety data in pregnant persons. May administer if at increased risk and vaccination benefits outweigh potential risks.
- For MenB recommendations **in outbreak setting** (eg, in community or organizational settings and among MSM) and **additional meningococcal vaccination information** refer to www.cdc.gov/mmwr/volumes/69/rr/rr6909a1.htm.
- MenB vaccines may be administered simultaneously with MenACWY vaccines if indicated, but at a different anatomic site, if feasible.
- Adults may receive a single dose of Penbraya (MenACWY-TT/MenB-FHbp) as an alternative to separate administration of MenACWY and MenB when both vaccines would be given on the same clinic day. For adults not at increased risk, if Penbraya is used for dose 1 MenB, then MenB-FHbp (Trumenba) should be administered for dose 2 MenB. For adults at increased risk of meningococcal disease, Penbraya may be used for additional MenACWY and MenB

doses (including booster doses) if both would be given on the same clinic day **and** at least 6 months have elapsed since most recent Penbraya dose.

▪ **Contraindications and precautions:**

- For contraindications and precautions to meningococcal ACWY (MenACWY) [MenACWY-CRM (Menveo); MenACWY-D (Menactra); MenACWY-TT (MenQuadfi)], refer to [MenACWY Appendix](#).
- For contraindications and precautions to meningococcal B (MenB) [MenB-4C (Bexsero); MenB-FHbp (Trumenba)], refer to [MenB Appendix](#).

¥¥ ***Haemophilus influenzae* type b vaccination**

▪ **Special situations:**

- **Anatomical or functional asplenia (including sickle cell disease):** 1 dose if previously did not receive Hib vaccine.
 - **Elective splenectomy:** 1 dose preferably at least 14 days before splenectomy.
- **Hematopoietic stem cell transplant (HSCT):** 3-dose series 4 weeks apart starting 6-12 months after successful transplant, regardless of Hib vaccination history.

▪ **Contraindications and precautions:**

- For contraindications and precautions to *Haemophilus influenzae* type b (Hib) vaccination, refer to [Hib Appendix](#).

‡‡ **Mpox vaccination**

▪ **Special situations:**

- **Any person at risk for mpox infection:** Complete 2-dose series, 28 days apart.

Risk factors for mpox infection include:

- Persons who are gay or bisexual and other MSM, transgender, or nonbinary people who in the past 6 months have had:
 - A new diagnosis of at least 1 sexually transmitted disease.
 - More than 1 sex partner.
 - Sex at a commercial sex venue.
 - Sex in association with a large public event in a geographic area where mpox transmission is occurring.
- Persons who are sexual partners of the persons described above.
- Persons who anticipate experiencing any of the situations described above.
- **Pregnancy:** There is currently no ACIP recommendation for Jynneos use in pregnancy due to lack of safety data in pregnant persons. Pregnant persons with any risk factor described above may receive Jynneos.
- **Health care personnel:** Vaccination to protect against occupational risk in health care settings is not routinely recommended.

▪ **Contraindications and precautions:**

- For contraindications and precautions to mpox, refer to [Mpox Appendix](#).

†† **Poliovirus vaccination**

▪ **Routine vaccination:**

- **Adults known or suspected to be unvaccinated or incompletely vaccinated:** Administer remaining doses (1, 2, or 3 inactivated poliovirus [IPV] doses) to complete a 3-dose primary series (NOTE: Complete primary series consists of at least 3 doses of IPV or trivalent oral poliovirus vaccine [tOPV] in any combination). Unless there are specific reasons to believe they were not vaccinated, most adults who were born and raised in the United States can assume they were vaccinated against polio as children.

▪ **Special situations:**

- **Adults at increased risk of exposure to poliovirus who completed primary series** (NOTE: Complete primary series consists of at least 3 doses of IPV or tOPV in any combination): May administer one lifetime IPV booster.

- For detailed information, refer to www.cdc.gov/vaccines/vpd/polio/hcp/recommendations.html.

- **Contraindications and precautions:**

- For contraindications and precautions to Poliovirus vaccine, IPV, refer to [Poliovirus Appendix](#).

*** Vaccinate after pregnancy, if indicated.

Reproduced from: Advisory Committee on Immunization Practices. Adult Immunization Schedule by Medical Condition and Other Indication, Recommendations for Ages 19 Years or Older, United States, 2025. Centers for Disease Control and Prevention. Available at: <https://www.cdc.gov/vaccines/hcp/imz-schedules/adult-medical-condition.html> (Accessed on December 10, 2024).

Graphic 62130 Version 27.0

Table 6: Indications for pneumococcal vaccination in adults in the United States

All adults ≥50 years of age
Adults 19 to 49 years of age with any of the following: ▪ Predisposing medical conditions: • Alcohol use disorder • Chronic heart disease[*] • Chronic lung disease[¶] • Chronic liver disease • Diabetes mellitus • Sickle cell disease or other hemoglobinopathies • Current cigarette smoking ▪ Increased risk of meningitis: • Cerebrospinal fluid leak • Cochlear implant ▪ Immunocompromising conditions and other conditions associated with altered immunocompetence^Δ: • Congenital or acquired immunodeficiency[◇] • Generalized active malignancy • HIV infection[§] • Iatrogenic immunosuppression[¥] • Hodgkin disease • Leukemia • Lymphoma • Multiple myeloma • Solid organ transplant • Chronic kidney disease[‡] and nephrotic syndrome ▪ Functional or anatomic asplenia ▪ History of invasive pneumococcal disease[†]

Pneumococcal vaccination is indicated for adults with risk factors for acquisition of or severe adverse outcomes from pneumococcal disease. These adults should receive PCV21 alone, PCV20 alone, or PCV15 followed by PPSV23. When administering the PCV15 and PPSV23 combination, PCV15 should be given first when possible. The recommended intervals between the 2 vaccines vary based on sequence and indication. Refer to the UpToDate topic on pneumococcal vaccination in adults for additional detail.

ACIP: Advisory Committee on Immunization Practices; HIV: human immunodeficiency virus; PCV15: 15-valent pneumococcal conjugate vaccine; PCV20: 20-valent pneumococcal conjugate vaccine; PCV21: 21-valent pneumococcal conjugate vaccine; PPSV23: 23-valent pneumococcal polysaccharide vaccine.

^{*} Including congestive heart failure and cardiomyopathies, excluding hypertension.

[¶] Including chronic obstructive pulmonary disease, emphysema, and asthma.

^Δ Some UpToDate authors differ from ACIP guidance on vaccine selection for immunocompromised individuals. Refer to the UpToDate topic on pneumococcal vaccination in adults for additional

information.

◇ Includes B (humoral) or T lymphocyte deficiency, complement deficiencies (particularly C1, C2, C3, and C4 deficiencies), and phagocytic disorders (excluding chronic granulomatous disease).

§ HIV infection is an indication for pneumococcal vaccination, regardless of CD4 cell count.

¥ Treatment with any immunosuppressive drugs (including long-term glucocorticoids, tumor necrosis factor alpha inhibitors, cancer chemotherapy, and other cytokine inhibitors) or radiation therapy.

‡ Chronic kidney disease is defined as glomerular filtration rate <60 mL/min/1.73 m² for ≥ 3 months.

† The United States Centers for Disease Control and Prevention ACIP does not mention individuals with a prior history of invasive pneumococcal disease in their recommendations on pneumococcal vaccinations. We suggest pneumococcal vaccination in this population due to their increased risk for recurrent pneumococcal disease. Refer to the UpToDate topic on pneumococcal vaccination in adults for additional detail.

References:

1. Kobayashi M, Pilishvili T, Farrar JL, et al. Pneumococcal vaccine for adults aged ≥ 19 years: Recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep* 2023; 72:1.
2. Centers for Disease Control and Prevention. CDC Newsroom: CDC Recommends Lowering the Age for Pneumococcal Vaccination from 65 to 50 Years Old. <https://www.cdc.gov/media/releases/2024/s1023-pneumococcal-vaccination.html> (Accessed January 15, 2026).

Table 7: Target groups for hepatitis B virus (HBV) vaccination in the United States

Patient groups	Comments
Infants (birth to 1 year)	▪ The vaccination series should be started as soon as possible after birth, preferably within 12 hours.* ▪ For infants born to mothers who are HBsAg positive, HBIG should be administered at the same time as the birth dose HBV vaccine (at a different anatomic site). ▪ For infants born to mothers who are HBsAg positive and mothers whose HBsAg status cannot be determined, obtain HBsAg and anti-HBs after completion of HBV vaccination series (usually at age 9 to 12 months). Revaccination may be necessary.
Unvaccinated persons age 1 to 60 years, regardless of risk for HBV infection	▪ Catch-up immunization is recommended for persons age <60 years who were not vaccinated for HBV or whose HBV vaccination status is unknown.
Unvaccinated persons age ≥60 years who are at increased risk for acquiring HBV or developing severe HBV infection as well as those who wish to be vaccinated This includes individuals with: ▪ Chronic liver disease ▪ HIV infection ▪ HCV infection ▪ Percutaneous or mucosal risk for HBV exposure (eg, injection drug use, occupational risk, household contact of someone with HBsAg) ▪ Sexual risk for HBV (eg, sexual contact with someone who is HBsAg positive, persons who are sexually active and not in mutually monogamous relationships) ▪ Planned travel to countries with high (≥8%) or intermediate (2 to 7%) endemic prevalence of HBV infection ▪ Risk due to being incarcerated ▪ Risk due to working or living in facilities for persons who are developmentally disabled ▪ Persons who are predialysis or are undergoing hemodialysis,	▪ Among individuals who are at increased risk for HBV, post-vaccination serologic testing (anti-HBs) is warranted for certain groups (eg, those with HIV, health care and public safety personnel, persons who are predialysis or are undergoing dialysis, and sexual partners of persons who are HBsAg positive).

This table lists target groups for HBV vaccination. More detailed information about groups at increased risk for HBV are presented in UpToDate content that discusses HBV epidemiology, screening, and immunization. For persons who are immunocompromised in all target groups, post-vaccination serology (anti-HBs) is recommended. Revaccination may be necessary. Refer to UpToDate content on HBV vaccination for details.

HBV: hepatitis B virus; HBsAg: hepatitis B surface antigen; HBIG: hepatitis B immune globulin; anti-HBs: antibody to HBsAg; HIV: human immunodeficiency virus; HCV: hepatitis C virus.

* Infants born to mothers who are HBsAg-positive and mothers whose HBsAg status cannot be determined should receive the first dose of HBV and HBIG as soon as possible after birth.

References:

1. Abara WE, Qaseem A, Schillie S, et al. Hepatitis B vaccination, screening, and linkage to care: Best practice advice from the American College of Physicians and the Centers for Disease Control and Prevention. *Ann Intern Med* 2017; 167:794.
2. Schillie S, Vellozzi C, Reingold A, et al. Prevention of hepatitis B virus infection in the United States: Recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep* 2018; 67:1.
3. Terrault NA, Lok AS, McMahon BJ, et al. Update on prevention, diagnosis, and treatment and of chronic hepatitis B: AASLD 2018 hepatitis B guidance. *Hepatology* 2018; 67:1560.
4. United States Preventive Services Task Force, Krist AH, Davidson KW, et al. Screening for hepatitis B virus infection in adolescents and adults: US Preventive Services Task Force Recommendation Statement. *JAMA* 2020; 324:2415.
5. Weng, MK, Doshani M, Khan MA, et al. Universal hepatitis B vaccination in adults aged 19-59 years: Updated recommendations from the Advisory Committee on Immunization Practices – United States, 2022. *MMWR Morb Mortal Wkly Rep* 2022; 71:477.
6. Conners EE, Panagiotakopoulos L, Hofmeister MG, et al. Screening and testing for hepatitis B virus infection: CDC recommendations – United States, 2023. *MMWR Recomm Rep* 2023; 72:1.

Table 8: Moderate to severe immunocompromising conditions that may result in suboptimal COVID-19 vaccine response

Active treatment for solid tumor and hematologic malignancies
Hematologic malignancies associated with poor responses to COVID-19 vaccines regardless of current treatment status (eg, chronic lymphocytic leukemia, non-Hodgkin lymphoma, multiple myeloma, acute leukemia)
Receipt of solid-organ transplant or an islet transplant and taking immunosuppressive therapy
Receipt of chimeric antigen receptor (CAR)-T cell therapy or hematopoietic cell transplant (HCT) (within 2 years of transplantation or taking immunosuppressive therapy)*
Moderate or severe primary immunodeficiency (eg, common variable immunodeficiency disease, severe combined immunodeficiency, DiGeorge syndrome, Wiskott-Aldrich syndrome)
Advanced HIV infection (HIV and CD4 cell counts less than 200/microL, history of an AIDS-defining illness without immune reconstitution, or clinical manifestations of symptomatic HIV) or untreated HIV infection
Active treatment with: ▪ High-dose corticosteroids (ie, ≥ 20 mg prednisone or equivalent per day for ≥ 2 weeks) ▪ Alkylating agents ▪ Antimetabolites ▪ Transplant-related immunosuppressive drugs ▪ Cancer chemotherapeutic agents classified as severely immunosuppressive ▪ TNF blockers ▪ Other biologic agents that are immunosuppressive or immunomodulatory (eg, B cell-depleting agents)

In the United States, the Centers for Disease Control and Prevention lists the above conditions as examples of immunocompromising conditions that warrant additional COVID-19 vaccine doses. This list is not exhaustive; other immunocompromising conditions, such as impaired splenic function, may also warrant the same vaccine adjustments. Refer to other UpToDate content for specifics of vaccine doses and intervals.

ACIP: Advisory Committee on Immunization Practices; CAR: chimeric antigen receptor; TNF: tumor necrosis factor.

* For those who received COVID-19 vaccination prior to hematopoietic stem cell transplant or CAR-T cell therapy, repeat vaccination is recommended at least 3 months after the transplant or therapy.

Reference:

1. COVID-19 Vaccination Guidance for People Who Are Immunocompromised. Centers for Disease Control and Prevention. Available at: https://www.cdc.gov/covid/hcp/vaccine-considerations/immunocompromised.html#cdc_cg_special_populations_section_3-description-of-moderate-and-severe-immunocompromising-conditions-and-treatment (Accessed on August 25, 2025).

Table 9: Health care considerations for sex workers

Clinical considerations	Comments
Cancer screening	Routine screening is performed for all cancers including those related to HPV (eg, cervical cancer screening).
Contraception	Provide information about contraception as indicated; the choice of contraceptive method is highly individualized and should be tailored to the individual's needs.
Documentation	- In settings where sex work is legal, a health check certificate may be requested by the patient. - In settings where sex work is illegal, documentation of sex work may warrant special consideration.
HIV PEP and PrEP	- Discuss PEP if a patient has a known or presumed exposure to HIV within the last 72 hours. - Discuss PrEP if the patient has ongoing risks of HIV exposure.
Immunizations	- Review immunization history. In particular, ensure appropriate vaccination for[*]: - HAV - HBV - HPV
STIs	- STI screening: - Chlamydia - Gonorrhea - Syphilis - Trichomonas (for those with a vagina) - HIV - HCV - HBV (vaccinate if not immune) The frequency of such screening depends on the assessed risk of exposure[¶]. Routine screening for herpes and/or genital warts is not performed (these are clinical diagnoses).
	- STI prevention: - Encourage male condom use^Δ. - Discuss PEP with doxycycline for selected individuals[◇].

Trauma, substance use, and mental health	▪ Regular screening for interpersonal violence, other trauma, and sex trafficking. ▪ Increased frequency of screening for substance use and mental health issues. ▪ Increased awareness of stigma encountered in medical care settings (especially in settings where sex work is criminalized).
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Sex workers, a term used to describe individuals who engage in sexual activity in exchange for money, material goods, or other services, are often not identified in the health care setting. However, such individuals may have specific healthcare needs; some of these considerations are included in this table.

HAV: hepatitis A virus; HBV: hepatitis B virus; HCV: hepatitis C virus; HIV: human immunodeficiency virus; HPV: human papillomavirus; PEP: postexposure prophylaxis; PrEP: pre-exposure prophylaxis; STIs: sexually transmitted infections.

* Other immunizations (eg, mpox vaccine) may be warranted in selected populations. Refer to related UpToDate content.

¶ In some locations where safer sex practices are commonplace among sex workers (through sex worker advocacy, peer-based education, support, and outreach services), rates of STIs among sex workers are similar to that of the general population. In such settings, annual screening may be reasonable. More frequent testing may be warranted in selected individuals (eg, individual prefers/requests more frequent testing, condom breaks, residing in a country with no/little safe sex practices among sex workers or where the legal status of sex work results in a lower likelihood of medical care).

Δ Condoms lubricated with spermicides are no more effective than other lubricated condoms in protecting against the transmission of HIV and other STIs.

◇ PEP with doxycycline can reduce the incidence of syphilis and chlamydia in selected populations. Refer to related UpToDate content.

Prepared with information from:

1. Sex workers. Australian STI Management Guidelines for Use in Primary Care. <https://sti.guidelines.org.au/populations-and-situations/sex-workers/> (Accessed on February 13, 2025).
2. Recommendations for evaluating donor eligibility using individual risk-based questions to reduce the risk of human immunodeficiency virus transmission by blood and blood products. US Food and Drug Administration. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/recommendations-evaluating-donor-eligibility-using-individual-risk-based-questions-reduce-risk-human> (Accessed on February 13, 2025).
3. McCann J, Crawford G, Hallett J. Sex Worker Health Outcomes in High-Income Countries of Varied Regulatory Environments: A Systematic Review. *Int J Environ Res Public Health* 2021; 18:3956.
4. American College of Obstetricians and Gynecologists. Improving awareness of and screening for health risks among sex workers. Committee Opinion No. 708. *Obstet Gynecol* 2017; 130:e53.

The editors of UpToDate would like to thank Jennefer Russo, MD, for her assistance in creating this table.

Table 10: Sexually transmitted infection screening recommendations by sex assigned at birth and population

Sex assigned at birth	Population	Routine screening recommendation	Screening frequency	Additional screening recommendations and comments
Cisgender females	Age <25 years and sexually active	Genital chlamydia [*]	Annually	If at increased risk [¶] , additionally screen for: ▪ Syphilis ▪ Trichomoniasis
		Genital gonorrhea [*]	Annually	
		HIV	At least once	
		HBV	At least once (if age ≥18 years and immunity not already documented) ^Δ	
		HCV	At least once (if age ≥18 years) [◇]	
	Age ≥25 years	HIV	At least once	If at increased risk [¶] , additionally screen for: ▪ Genital chlamydia and gonorrhea[*] ▪ Syphilis ▪ Trichomoniasis
		HBV	At least once (if immunity not already documented) ^Δ	
		HCV	At least once [◇]	
	Pregnant	Genital chlamydia [*]	First trimester (if <25 years or at increased risk [¶])	Repeat screening for these infections in third trimester if at increased risk. Additional screening at first prenatal visit: ▪ HCV for those at risk (or if ≥18 years with no prior screening)[◇] ▪ Trichomoniasis for those with HIV
		Genital gonorrhea [*]	First trimester (if <25 years or at increased risk [¶])	
		Syphilis	First trimester, third trimester, and at birth	
		HIV	First trimester	
		HBV	First trimester	

	With HIV infection	Genital chlamydia [*]	Annually	
		Genital gonorrhea [*]	Annually	
		Genital trichomoniasis	Annually	
		Syphilis	Annually	
		HBV	At least once (eg, at first visit)	
		HCV	At least once (eg, at first visit) [◇]	
	WSW and WSWM	WSW and WSWM should not be assumed to be at lower risk for STIs on the basis of their sexual orientation. Screening for cervical cancer and STIs should be conducted according to guidelines for women, based on an open discussion of sexual and behavioral risk factors.		
Cisgender males	MSW only without HIV infection	HIV	At least once	If at increased risk [§] , additionally screen for: ▪ Genital chlamydia and gonorrhea ▪ Syphilis Targeted screening venues for chlamydia include adolescent clinics, STI clinics, and correctional facilities.
		HBV	At least once (if age ≥ 18 years and immunity not already documented) ^Δ	
		HCV	At least once (if age ≥ 18 years) [◇]	
	MSM without HIV infection	Genital chlamydia	At least annually	More frequent screening (every 3 to 6 months) for chlamydia, gonorrhea, and syphilis is recommended in those with risk factors, including those using PrEP or doxy PEP. More frequent screening for HIV,
		Rectal chlamydia (if exposed)	At least annually	
		Genital gonorrhea	At least annually	

		Rectal gonorrhea (if exposed)	At least annually	HBV, and HCV may also be warranted.¥
		Pharyngeal gonorrhea (if exposed)	At least annually	
		Syphilis	At least annually	
		HIV	At least annually	
		HAV	At least once	
		HBV	At least once	
		HCV	At least once◇	
	MSW only with HIV infection	Genital chlamydia	Annually	
		Genital gonorrhea	Annually	
		Syphilis	Annually	
		HBV	At least once (eg, at first visit)	
		HCV	At least once (eg, at first visit)◇	
	MSM with HIV infection	Genital chlamydia	At least annually	More frequent screening (every 3 to 6 months) for chlamydia, gonorrhea, and syphilis is recommended in those with risk factors, including those using doxy PEP. More frequent screening for HBV and HCV may also be warranted.¥
		Rectal chlamydia (if exposed)	At least annually	
		Genital gonorrhea	At least annually	
		Rectal gonorrhea (if exposed)	At least annually	
		Pharyngeal gonorrhea (if exposed)	At least annually	
		Syphilis	At least annually	
		HAV	At least once (eg, at first visit)	
		HBV	At least once (eg, at first visit)	

		HCV	At least annually [◇]	
Transgender and gender-diverse individuals		Screening for STIs should be based on an individual's anatomy: ▪ Transgender men and gender-diverse individuals with a cervix should be screened for genital gonorrhea, chlamydia, and cervical cancer according to recommendations for cisgender women. ▪ Transgender women and gender-diverse individuals assigned male at birth should be screened for genital gonorrhea and chlamydia according to recommendations for cisgender men. Screening for other STIs should be based on sexual practice, risk factors, and exposures.		

doxy PEP: doxycycline postexposure prophylaxis; HAV: hepatitis A virus; HBV: hepatitis B virus; HCV: hepatitis C virus; HIV: human immunodeficiency virus; MSM: men who have sex with men; MSW: men who have sex only with women; PrEP: pre-exposure prophylaxis; STI: sexually transmitted infection; WSW: women who have sex with women; WSWM: women who have sex with women and men.

* Screening for nongenital infections in females (eg, rectal chlamydial infection, pharyngeal and rectal gonococcal infection) can be considered based on reported sexual behaviors and exposure, via shared clinical decision-making between the patient and the provider.

¶ Factors conferring increased risk for gonorrhea, chlamydia, and trichomoniasis in females include transactional sex, new sex partner, multiple sex partners, a sex partner with concurrent partners, or a sex partner with an STI. Risk factors for syphilis include residence in [high-prevalence areas](#), history of incarceration, or transactional sex work. STI screening may also be considered in high-prevalence settings (eg, STI clinic or correctional facility).

Δ For all adults 18 years of age or older, regardless of risk factors, at least 1-time screening for HBV infection is recommended, unless they have documented vaccine receipt and serologic evidence of vaccine response. Those who are susceptible should be vaccinated. For those who have risk factors for HBV exposure, ongoing screening is warranted if they are unvaccinated or have nonresponse to vaccination. Refer to other UpToDate content on STIs for details.

◇ All adults 18 years of age or older should be screened for HCV at least once, except in settings where the HCV positivity is <0.1%. Repeated screening is warranted for those with ongoing risk factors (eg, injection drug use). Increased risk factors for hepatitis C infection among MSM include HIV infection, high community HCV prevalence and incidence, high-risk sexual behaviors, and concomitant ulcerative STIs or STI-related proctitis. Refer to other UpToDate content on hepatitis C screening for details.

§ Factors conferring increased risk for gonorrhea and chlamydia in MSW include an infection in the preceding 24 months. Screening for chlamydia in young males can be considered in high-prevalence clinical settings (adolescent clinics, correctional facilities, STI/sexual health clinics). Increased risk factors for syphilis may be based on geography, race/ethnicity, history of incarceration, transactional sex work, or age <29 years.

¥ Increased risk factors for gonorrhea, chlamydia, syphilis, and HIV among MSM include multiple or anonymous partners; intravenous drug use; sex in conjunction with illicit drug use, including methamphetamines; and sex partners who engage in these activities. MSM who have not been vaccinated for HBV or have had nonresponse to vaccination remain at risk for HBV infection.

1. California sexually transmitted infections (STI) screening recommendations, 2021. California Department of Public Health. <https://www.cdph.ca.gov/Programs/CID/DCDC/Pages/California-STI-Screening-Recommendations.aspx> (Accessed on January 24, 2023).
 2. Workowski KA, Bachmann LH, Chan PA, et al. sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep* 2021; 70:1.
 3. Conners EE, Panagiotakopoulos L, Hofmeister MG, et al. Screening and testing for hepatitis B virus infection: CDC recommendations – United States, 2023. *MMWR Recomm Rep* 2023; 72:1.
 4. Screening for syphilis in pregnancy. American College of Obstetricians and Gynecologists. <https://www.acog.org/clinical/clinical-guidance/practice-advisory/articles/2024/04/screening-for-syphilis-in-pregnancy> (Accessed on November 11, 2025).
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Table 11: **Groups at increased risk for hepatitis B virus infection**

Individuals at risk for HBV due to vertical transmission (ie, mother-to-child transmission) ▪ Individuals born in regions with high ($\geq 8\%$) or intermediate ($\geq 2\%$) prevalence rates for HBV, including immigrants and adopted children[*] ▪ Infants born to pregnant persons who are HBsAg positive[¶] ▪ US-born persons not vaccinated as infants whose parents were born in regions with high HBV endemicity ($\geq 8\%$)[*]
Individuals at risk due to horizontal transmission (ie, percutaneous or mucosal exposure to blood or body fluids contaminated with blood)^Δ ▪ Household contacts of HBsAg-positive persons ▪ Needle sharing or sexual contacts of HBsAg-positive persons ▪ Individuals who have ever injected drugs ▪ Individuals with multiple sexual partners and/or history of sexually transmitted infections ▪ Men who have sex with men ▪ Inmates of correctional facilities or other detention settings ▪ Individuals with HIV infection[◇] ▪ Individuals with current or past HCV infection[§] ▪ Individuals with end-stage kidney disease on maintenance renal dialysis
Other individuals ▪ Individuals with elevated alanine aminotransferase or aspartate aminotransferase levels of unknown etiology ▪ Individuals who request HBV testing

In the United States, screening for HBV includes^[1]:

- **Risk-based screening** – For all individuals (including children and adolescents), screen those who have any of the risk factors listed in the table if they might have been susceptible during the period of increased risk[¥]. For those with ongoing risk factors (ie, for horizontal transmission) who remain susceptible, continue to test periodically.^Δ
- **Universal screening** – For individuals ≥ 18 years of age, screen at least once in a lifetime. However, for those without risk factors for HBV, screening is generally not needed if there is documentation of completing a hepatitis B vaccine series and evidence of immunity (anti-HBs ≥ 10 milli-international units/mL) after vaccination.[‡]
- **Pregnancy screening** – Screen all pregnant people during each pregnancy, regardless of vaccination status or history of prior testing.

Refer to UpToDate content on screening and diagnosis of HBV, HBV immunization, and HBV and pregnancy for more detailed information on screening and vaccination.

anti-HBc: hepatitis B core antibodies; anti-HBs: hepatitis B surface antibody; HBIG: hepatitis B immune globulin; HBsAg: hepatitis B surface antigen; HBV: hepatitis B virus; HCV: hepatitis C virus; HIV: human immunodeficiency virus; US: United States.

* If HBsAg-positive persons are found in first-generation immigrants of a family, subsequent generations should be tested.

¶ To reduce the risk of perinatal transmission, infants born to HBsAg-positive mothers should receive HBIG and hepatitis B vaccine as soon as possible and within 12 hours of birth and then complete the hepatitis B series. Postvaccination serology should be obtained at 9 to 12 months. Refer to the UpToDate topic that discusses HBV immunization in infants.

Δ In unvaccinated individuals with ongoing HBV risk through percutaneous or mucosal exposure, hepatitis B vaccine should be initiated at the time of screening; the need for subsequent doses will depend upon the results. Postvaccination serology should be performed to ensure immunity. For at-risk persons who do not complete the vaccine series, repeat testing should be performed periodically (eg, every 1 to 2 years).

◇ The presence of HBV coinfection informs the choice of antiretroviral regimen. In addition, patients with HIV who are not immune should be vaccinated regardless of age or risk factors, since HBV infection has an accelerated course in coinfecting patients.

§ Patients with chronic HBV are at risk for HBV reactivation with direct-acting antiviral therapy for hepatitis C. Refer to the UpToDate topic that provides an overview of the management of hepatitis C infection.

¥ Susceptible persons include those who have never been infected with HBV (ie, HBsAg-negative, total anti-HBc-negative, and anti-HBs-negative) and either did not complete a hepatitis B vaccine series per the Advisory Committee on Immunization Practices recommendations or who are known to be vaccine nonresponders.

‡ For most patients who remain without risk factors for acquiring HBV, repeat screening is not warranted. However, screening prior to blood, plasma, organ, tissue, or semen donation is routinely performed, regardless of the person's prior history. In addition, screening is warranted prior to initiating immunosuppressive therapy (eg, corticosteroids, biologics, cancer chemotherapy, antirejection therapies) since persons with HBV are at risk for HBV reactivation. Refer to the UpToDate topic on HBV reactivation.

Reference:

1. Conners EE, Panagiotakopoulos L, Hofmeister MG, et al. Screening and testing for hepatitis B virus infection: CDC recommendations – United States, 2023. *MMWR Recomm Rep* 2023; 72:1.

With additional information from:

- Weinbaum CM, Williams I, Mast EE, et al. Recommendations for identification and public health management of persons with chronic hepatitis B virus infection. *MMWR Recomm Rep* 2008; 57:1.
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Table 12: Short Patient Health Questionnaire (PHQ-2)

Name:		Date:			
Over the past 2 weeks, how often have you been bothered by any of the following problems?	Not at all	Several days	More than half the days	Nearly every day	
Little interest or pleasure in doing things?	0	1	2	3	
Feeling down, depressed, or hopeless?	0	1	2	3	
Total point score:	—	—	+ —	+ —	+ —

Score interpretation^[1]:		
PHQ-2 score	Probability of major depressive disorder (%)	Probability of any depressive disorder (%)
1	15.4	36.9
2	21.1	48.3
3	38.4	75.0
4	45.5	81.2
5	56.4	84.6
6	78.6	92.9

Reference:

1. Kroenke K, Spitzer RL, Williams JB. The Patient Health Questionnaire-2: validity of a two-item depression screener. *Med Care* 2003; 41:1284.

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Table 13: PHQ-9 depression questionnaire

Name:	Date:			
Over the last 2 weeks, how often have you been bothered by any of the following problems?	Not at all	Several days	More than half the days	Nearly every day
Little interest or pleasure in doing things	0	1	2	3
Feeling down, depressed, or hopeless	0	1	2	3
Trouble falling or staying asleep or sleeping too much	0	1	2	3
Feeling tired or having little energy	0	1	2	3
Poor appetite or overeating	0	1	2	3
Feeling bad about yourself, that you are a failure, or that you have let yourself or your family down	0	1	2	3
Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
Moving or speaking so slowly that other people could have noticed; or the opposite, being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3
Total ___ =	___	+ ___	+ ___	+ ___
PHQ-9 score ≥ 10: Likely major depression				
Depression score ranges:				
5 to 9: mild				
10 to 14: moderate				
15 to 19: moderately severe				
≥ 20 : severe				
If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?	Not difficult at all ___	Somewhat difficult ___	Very difficult ___	Extremely difficult ___

PHQ: Patient Health Questionnaire.

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Table 14: AUDIT-C

Question #1: How often did you have a drink containing alcohol in the past year?	
• Never	(0 points)
• Monthly or less	(1 point)
• 2 to 4 times a month	(2 points)
• 2 to 3 times per week	(3 points)
• 4 or more times a week	(4 points)
Question #2: How many drinks did you have on a typical day when you were drinking in the past year?	
• 1 or 2	(0 points)
• 3 or 4	(1 point)
• 5 or 6	(2 points)
• 7 to 9	(3 points)
• 10 or more	(4 points)
Question #3: How often did you have 6 or more drinks on one occasion in the past year?	
• Never	(0 points)
• Less than monthly	(1 point)
• Monthly	(2 points)
• Weekly	(3 points)
• Daily or almost daily	(4 points)

The AUDIT-C is scored on a scale of 0 to 12 (scores of 0 reflect no alcohol use). In men, a score of 4 or more is considered positive; in women, a score of 3 or more is considered positive.

Table 15: Clinical risk factors for fracture

Advanced age (eg, ≥ 65 years for women)*
Previous fracture during adulthood
High fall risk
Impaired mobility
Chronic glucocorticoid therapy (eg, ≥ 3 months)
Parental history of hip fracture
Low body weight (<58 kg [127 lb])
Current cigarette smoking
Excess alcohol intake
Height loss (>3.8 cm [1.5 in]) and/or kyphosis
Secondary osteoporosis (eg, rheumatoid arthritis, hyperthyroidism, primary hyperparathyroidism, hypogonadal states, gastrointestinal diseases and malabsorptive states)

BMD: bone mineral density.

* Most UpToDate contributors do not routinely measure BMD in men based on age alone. Instead, they perform targeted testing in men aged ≥ 50 years with ≥ 1 clinical risk factor for fracture. However, some experts suggest measuring BMD in men aged ≥ 70 years, irrespective of other clinical risk factors.

Reference:

1. Kanis JA, Borgstrom F, De Laet C, et al. Assessment of fracture risk. *Osteoporos Int* 2005; 16:581.
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